



Investor sentiment and sovereign bonds

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ABSTRACT

Investor sentiment is an important driver of sovereign bond returns in emerging markets. Using local sovereign debt and external sovereign debt, this paper finds that sentiment is negatively related to future bond returns on average across emerging market countries. This negative sentiment effect suggests that investors treat emerging market sovereign bonds as risky assets rather than as safe assets. This paper further finds that this sentiment effect is relatively stronger when liquidity frictions are higher, which is consistent with illiquid bonds being generally harder to arbitrage.

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1. Introduction

A number of papers have examined which empirical asset pricing models best explain international government bond returns. For instance, [Cochrane and Piazzesi \(2005\)](#) find that a single linear combination of forward rates explains between 31% and 37% of excess bond returns. [Ludvigson and Ng \(2009\)](#) construct fitted factors from 132 monthly economic series using dynamic factor analysis and use these common factors to forecast future excess returns on US government bonds.¹ However, these widely used predictors do not consider any role for behavioral factors (e.g. investor sentiment) in explaining bond returns. This paper takes advantage of a large panel data set of emerging market (EM) sovereign bond returns to test the relationship between US equity market investor sentiment and future sovereign bond returns. Sentiment is shown to have a significant effect on emerging market sovereign bond returns, and this effect is larger for less liquid securities.

As early as 1936, Keynes proposes that investors' "animal spirits" move market prices in way unrelated to fundamentals ([Keynes, 1936](#)). Studies have found persistent deviations from efficient prices and investors are subject to these "animal spirits"- sentiment ([De Long, Shleifer, Summers, and Waldman, 1990](#); [Lee, Shleifer, and Thaler, 1991](#)). Investor sentiment, defined broadly, captures beliefs about asset price deviations from present value of future cash flows. [Baker and Wurgler \(2006, 2007\)](#) construct an investor sentiment index to capture the level of market-wide mispricing in the stock market. In the past two decades, mounting research links stock market sentiment to expected equity returns ([Brown and Cliff, 2004](#); [Schmeling, 2009](#); [Yu and Yuan, 2011](#); [Shen, Yu, and Zhao, 2017](#); [Gao et al., 2020](#)). Recently, a literature also documents that the effect of sentiment on stock prices spills over to bonds ([Bethke, Gehde-Trapp, and Kempf, 2017](#)). [Barberis, Shleifer,](#)

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¹ Scholars have proposed various factors or predictors to explain bond returns. [Fama and French \(1993\)](#) show that term spread and default spread capture much of the variation in bond returns, and [Harvey, Solnik, and Zhou \(2002\)](#) find that the factors driving bond returns are the same as those driving equity returns. [Perignon, Smith, and Villa \(2007\)](#) use a principal component analysis to investigate the common factor structure among US, German, and Japanese Government bond returns.

and Wurgler (2005) argue that bond returns comove with world markets based on news on fundamentals and investor sentiment. Baker and Wurgler (2012) find that bond-like stocks and bonds have relatively high returns when sentiment is high.

This study is mainly motivated by Baker and Wurgler (2006)'s finding that, when investor sentiment is low, future returns are relatively high for small stocks, high volatility stocks, and distressed stocks. There are limits to arbitrage when arbitrageurs bet against sentiment-driven mispricings of these stocks (Shleifer and Vishny, 1997). Empirical research shows that these stocks are costly to trade and difficult to sell short (Amihud and Mendelson, 1986; Stambaugh, Yu, and Yuan, 2012). Moreover, these stocks tend to have high idiosyncratic risk, which makes them relatively hard to arbitrage (Pontiff, 2006). Their illiquidity also results in limits to arbitrage (Brunnermeier and Pedersen, 2005). Additionally, as Erb, Harvey, and Viskanta (1999) show, during the 1990's, emerging market bonds had high yields, high volatility, and were highly correlated with equity markets. Thus, EM sovereign bonds are more equity-like assets than safe assets. Currently, EM sovereign bonds are treated as a separate asset class rather than as safe assets, and this categorization continues even after the globalization of past two decades (Bekaert and Harvey, 2017; Caballero, Farhi, and Gourinchas, 2017).

Given the significant body of empirical work documenting investor sentiment and arbitrage, and EM sovereign bonds' equity-like characteristics, this paper tests several hypotheses. First, this paper tests whether US sentiment has positive or negative effects on the subsequent returns of EM sovereign bonds. Second, this paper tests whether sentiment effects vary among EM local debt and external (USD) debt. Third, this study investigates whether bond liquidity affects the sentiment-return relation for EM debt.

Several papers focus on the relationship between sentiment and US bond markets: Baker and Wurgler (2012) and Laborda and Olmo (2014) find a positive relation between sentiment and bond risk premia; Cepni, Guney, Gupta and Wohar (2020) construct an aligned investor sentiment index using the partial least squares approach and use the index to predict the excess returns of US government bonds; Guo, Lin, Wu, and Zhou (2019) propose a bond sentiment index equal to the difference between the observed yield spread and model fitted yield spread, and use it to predict US corporate bond returns; Chiu, Harris, Stoja, and Chin (2018) examine the relationship between sentiment and US stock and bond volatility; other studies explore sentiment effects in the US corporate bond markets (Nayak, 2010; Huang, Rossi, and Wang, 2015; Li and Galvani, 2018). However, not many papers assess the impact of investor sentiment on EM sovereign bonds.² Over the last three decades, emerging markets play an increasing role in the world economy. Fig. 1 displays the relative share of world GDP for emerging markets (EM) and developed markets (DM) in the sample. In 1985, EM GDP represents only 30% of world GDP, whereas after 2015 EM GDP accounts for more than 56.5% of world GDP. As Fig. 2 indicates, EM sovereign debt markets are an increasingly important portion of total sovereign financing. Global investors, such as asset managers, mutual funds, and institutional investors, increase their amount of EM sovereign debt holding recently (Miyajima and Shim, 2014). Previous studies of sovereign debt in EM countries mainly focus on external debt because of "original sin," the inability of EM countries to borrow abroad in their domestic currency (Eichengreen and Hausmann, 1999; Tirole, 2003; Eichengreen et al., 2003).³ After 2000, many EM countries started to issue local currency debt and retired their external debt (IMF, 2002, 2005). There has also been an increase in the foreign ownership of EM local currency bonds over the past decade (Miyajima, Mohanty, and Chan, 2015). Because of increasing local currency denominated debt issuance and the importance of foreign currency debt in EMs, I consider both local debt and external (USD) debt in the analysis. The sample consists of local debt and external (USD) debt data from 30 EM countries and covers the period from January 1998 to September 2015.

Following the literature, this article primarily adopts the investor sentiment index proposed by Baker and Wurgler (2006) to explore sentiment effects. With globalization, global shocks appear to have large effects on the returns of EM assets (Eichengreen and Mody, 2000; IMF, 2004). The US plays a particularly dominant role in affecting global markets.⁴ Furthermore, US investor sentiment is highly correlated with global sentiment (Baker, Wurgler, and Yuan, 2012). Consistent with the hypothesis, I find that US sentiment has an impact on EM sovereign bonds. Understanding the drivers of EM sovereign bond returns provides insights into sovereign asset pricing. Changes in EM sovereign bond returns may trigger reweighting of global debt portfolios, which would further affect the cost and the flow of international capital. The analysis of sovereign bond pricing is important for bond portfolio management, risk management and regulation of financial institutions. In this respect, the paper contributes a better understanding of EM sovereign bond prices.

The main contribution of this study is to document a negative relationship between investor sentiment and future excess sovereign bond returns for emerging markets. This relationship is maintained after controlling for a tent-shaped linear combination of forward rates, benchmark macroeconomic factors, the Fama-French six factors, well-known global economic variables, and other local macro variables. Economically, a one standard deviation increase in sentiment is associated with a 43.8 basis point decrease in EM local sovereign bond returns and a 50.8 basis point decrease in external (USD) sovereign bond returns.

High sentiment indicates a bullish stock market and investors' optimism about stocks in general. During high sentiment periods, investors are more optimistic about risky assets and EM bond markets are overpriced. EM sovereign bonds are

² Some studies investigate the relationship between market sentiment and sovereign bond yields in the Euro area (Spyrou, 2013; Aristei and Martelli, 2014). Pineiro-Chousa et al. (2021) explore how the investor sentiment extracted from a social network affects the green bond market.

³ Eichengreen et al., 2003 suggest that "original sin" causes a potential mismatch between the cash flows generated by the project and the borrowing currency. EM governments may also use foreign currency borrowing to strengthen their international reserves.

⁴ Many studies focus on various shocks originating from the US: monetary policy shocks (Mackowiak, 2007), stock market shocks (Soydemir, 2000), and sentiment shocks (Schmeling, 2009). Longstaff (2010) even argues how the US subprime crisis spilled over and grew into a global financial crisis.

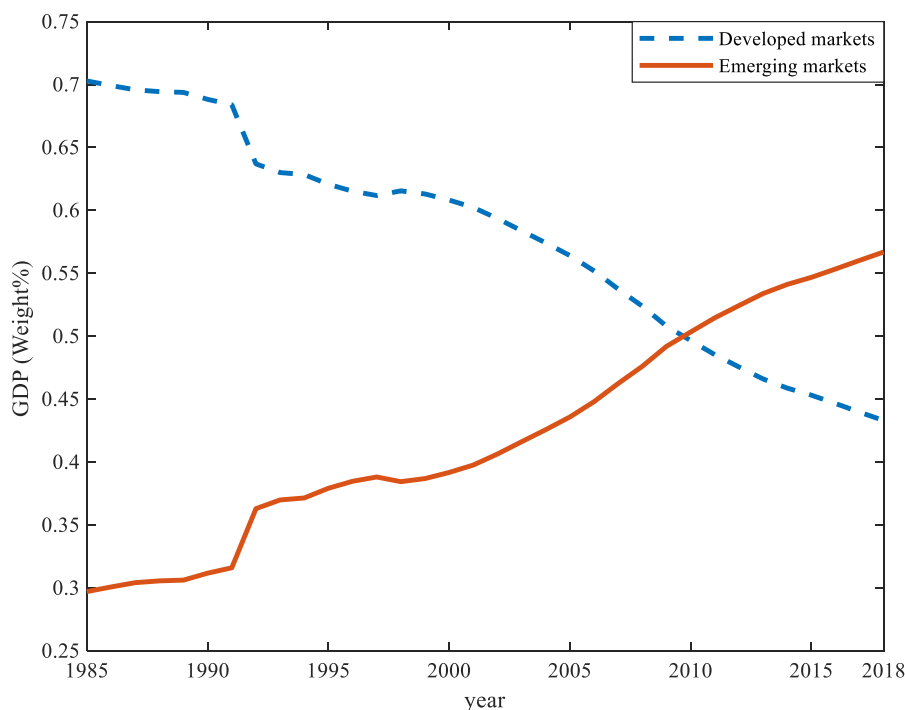


Fig. 1. Emerging and Developed Countries' Share of World GDP. This figure shows the share of world GDP for developed countries and emerging markets over the period 1985–2018. Emerging markets (EM) in our sample include countries: Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Israel, Latvia, Lithuania, Malaysia, Mexico, Morocco, Nigeria, Norway, Pakistan, Peru, Philippines, Poland, Romania, Russia, Slovak Republic, Slovenia, South Africa, South Korea, Sri Lanka, Thailand, Turkey, and Vietnam. Developed markets (DM) in our sample include countries: Australia, Austria, Belgium, Canada, Cyprus, Denmark, Finland, France, Germany, Hong Kong, Ireland, Israel, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, Taiwan, UK, and US. Source: IMF World Economic Outlook 2018. GDP data in US dollars.

generally considered to be riskier, equity-like assets, while sovereign bonds from developed countries are considered to be safe assets. When investor sentiment declines, investors withdraw from riskier assets and move to safer sovereign bonds. This flight-to-safety leads to negative sentiment effects on EM sovereign bond returns but positive sentiment effects on safe assets (sovereign bonds from developed markets). Consistent with previous studies (e.g., [Baker and Wurgler, 2012](#); [Laborda and Olmo, 2014](#)), this study also finds that investor sentiment is positively related with future returns of sovereign bonds from developed markets (DM). Overall, this paper finds that low (high) investor sentiment periods are related to greater (lower) sovereign bond returns in emerging markets, and lower (greater) sovereign bond returns in developed markets. Consistent with [Bekaert and Harvey \(2017\)](#), the results indicate that EMs are not fully integrated with global markets and that EM sovereign bonds should be treated as a separate asset class.

Additionally, this article explores how EM sovereign bond liquidity affects this sentiment–return relationship. Using the bid–ask spread to proxy for EM sovereign bond liquidity, I find that sentiment effects are stronger for EM sovereign bonds that are more illiquid due to limits of arbitrage. The results suggest that the illiquid EM local debt is more vulnerable to sentiment shocks than illiquid EM external debt.

This paper is most closely related to [Weigel and Gemmill \(2006\)](#). They investigate how global and local factors impact creditworthiness of four EMs. They fit a structural model with a Kalman filter to Brady bonds to estimate distance to default, a measure of creditworthiness and use the discounts on closed-end funds to proxy for regional sentiment. Their sentiment analysis includes Argentina, Brazil, and Mexico from April 1994 to October 2001. My paper differs from theirs in several important respects. First, my study focuses on the relationship between US sentiment and future EM sovereign bond returns. This paper uses the Baker and Wurgler sentiment index, which is the first principal component of the correlation matrix of six measures of investor sentiment. My research approach focuses on investors' perspective of EM sovereign bond index returns than yields. Bond returns are the holding period yields when investors trade the bonds before maturity. Bond returns include accrued or received coupon, amortization payments, and capital gains. Second, my study uses a dynamic panel data model with extensive sample coverage of 30 EMs over a more recent sample period, from 1998 to 2015. The advantage of using this longer sample period is that it covers peaks and troughs of two business cycles. Furthermore, this paper considers the sovereign bond returns for both local debt and external (USD) debt.

The rest of the paper is organized as follows. [Section 2](#) describes the data and sources. [Section 3](#) presents some facts of sovereign bonds for emerging economies and presents empirical results of investor sentiment on sovereign bond returns. [Section 4](#) discusses liquidity's impact on sentiment–return relationship. [Section 5](#) presents robustness check analyses. [Section 6](#) summarizes the results and presents concluding remarks.

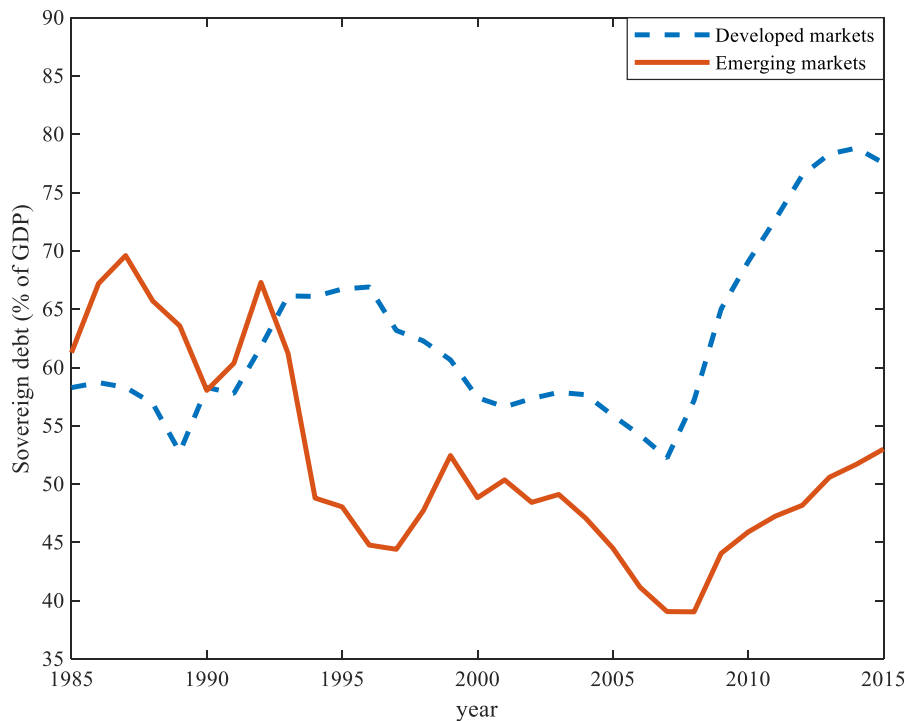


Fig. 2. Emerging and Developed Countries' Sovereign Debt. This figure shows sovereign debt (% GDP) for developed countries and emerging markets over the period 1985–2015. Emerging markets (EM) in my sample include countries: Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Israel, Latvia, Lithuania, Malaysia, Mexico, Morocco, Nigeria, Norway, Pakistan, Peru, Philippines, Poland, Romania, Russia, Slovak Republic, Slovenia, South Africa, South Korea, Sri Lanka, Thailand, Turkey, and Vietnam. Developed markets (DM) in my sample include countries: Australia, Austria, Belgium, Canada, Cyprus, Denmark, Finland, France, Germany, Hong Kong, Ireland, Israel, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, Taiwan, UK, and US. Source: IMF World Economic Outlook 2018.

2. Data and sources

2.1. Sentiment measures

This paper adopts the investor sentiment index developed by Baker and Wurgler (2006) (hereafter, the BW sentiment index) available at Professor Jeffrey Wurgler's web page (<http://people.stern.nyu.edu/jwurgler/>). The monthly BW sentiment index spans over 45 years, from July 1965 to September 2015. In this study, I consider the sample from January 1998 to September 2015 due to the availability of sovereign bond data for emerging economies (see Section 2.2).

Baker and Wurgler (2006) construct their investor sentiment composite index by taking the first principal component of the correlation matrix of six measures of investor sentiment. The six measures are the closed-end fund discount, the number and the first-day returns of initial public offerings (IPOs), the turnover of stocks traded on the New York Stock Exchange (NYSE), the share of equity issues in total equity and debt issues, and the dividend premium. This principal component, denoted as S_t^{BW} in this paper, filters out idiosyncratic noise in the six measures and captures their common component. Furthermore, they orthogonalize the six measures with respect to six macroeconomic indicators: growth in industrial production; real growth in durable, nondurable, and services consumption; growth in employment; and the National Bureau of Economic Research (NBER) recession indicator. I denote the first principal component of the six orthogonalized proxies as $S_t^{BW\perp}$ in this paper.

Fig. 3 shows the time series plot of the BW sentiment index S_t^{BW} and $S_t^{BW\perp}$. The BW sentiment index S_t^{BW} went up before 2000 and fell after the Internet bubble. It then picked up and reached a peak before the recent financial crisis. The orthogonal BW sentiment index $S_t^{BW\perp}$ fluctuates more as it is orthogonal to market conditions.

2.2. Sovereign bond data

This study explores whether investor sentiment can predict EM sovereign bond returns. Recent trends highlight the importance of local currency debt. After 2000, many EMs choose to substitute local debt for external debt. Fig. 4 shows the increase in the market value of local currency sovereign debt over the past two decades.

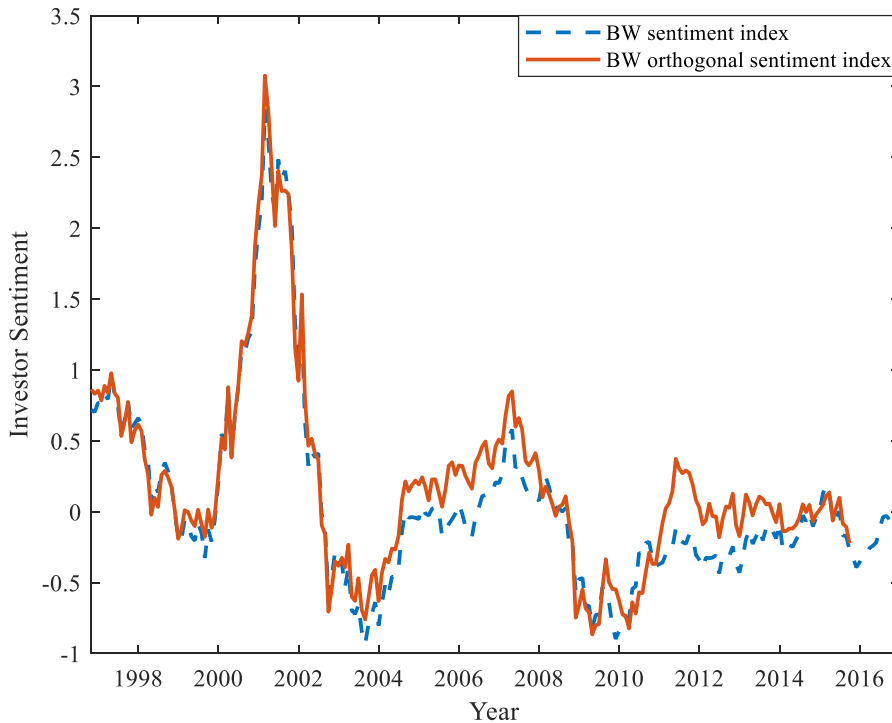


Fig. 3. The Investor Sentiment Index From 1998 to 2015. This figure displays the time series plot of investor sentiment series by Baker and Wurgler (2006): BW sentiment index S_t and BW sentiment index S_t^+ . The series in solid line is the first principal component of the six measures. The six measures are the closed-end fund discount, the number and the first-day returns of initial public offerings (IPOs), turnover of stocks traded on New York Stock Exchange (NYSE), the share of equity issues in total equity and debt issues, and the dividend premium. The BW sentiment series in dashed line is the one where each of the six measures has first been orthogonalized with respect to a set of macroeconomic conditions.

I use the ICE Bank of America Merrill Lynch Government Total Return Indexes denominated in local currency and the ICE Bank of America Merrill Lynch US Dollar Emerging Markets External Sovereign Indexes from Bloomberg to calculate sovereign bond returns.⁵ The sample consists of monthly data from 30 EM countries and covers the period from January 1998 to October 2018. Specifically, I study the following 30 EMs: Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Latvia, Lithuania, Malaysia, Mexico, Morocco, Nigeria, Pakistan, Peru, Philippines, Poland, Romania, Russia, Slovak Republic, Slovenia, South Africa, South Korea, Sri Lanka, Thailand, Turkey, and Vietnam.⁶ Table 1 reports the inception year and basic statistics for these EM sovereign bond indexes. The data from the Czech Republic, Greece, Hungary, Poland, and South Africa have the earliest start date in 1998. For local sovereign debt, China has the highest average full market value; for external (USD) debt, Brazil has the highest average full market value. Chile has the highest average credit rating for both local debt and external (USD) debt, while Pakistan has the lowest average credit rating for both types of debt.

Each country's index is market capitalization-weighted. Securities in each index have at least 18 months to maturity at issuance, and at least one year remaining term to maturity. Sovereign bond USD returns for emerging economies are measured in excess of the U.S. 1-month Treasury bill returns from Ibbotson Associates if the returns are in the US dollar. To compute the excess return valued in the local currency, I subtract the local currency risk-free rate.⁷ The control factors used in this set of tests are in Section 2.3 and Section 2.4.

In Section 3.4, I analyze the effects of investor sentiment on the sovereign bond returns from developed countries as a comparison analysis to EMs. The developed countries are: Australia, Austria, Belgium, Canada, Cyprus, Denmark, Finland, France, Germany, Hong Kong, Ireland, Israel, Italy, Japan, Luxembourg, Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, Taiwan, UK, and US.⁸

⁵ Like prior research, this paper takes the perspective of a US investor and measure sovereign bond returns valued in US dollar as well for both local debt and external (USD) debt. For local debt, I also report the analyses of bond returns valued in the local currency.

⁶ The sample of EMs in this paper is based on the MSCI market classification and constituents of EM sovereign bond indexes of ICE Bank of America Merrill Lynch.

⁷ This paper uses the interbank rate or overnight rate to proxy for a risk-free rate if country's short-term treasury yield is not available in Datastream.

⁸ Previous literature mainly focuses on US bond markets or one specific European country. Baker and Wurgler (2012) find that a positive relationship between sentiment and excess returns of US bond on different maturities. This article includes 26 countries in developed market analysis and use the excess returns of their sovereign bond indexes.

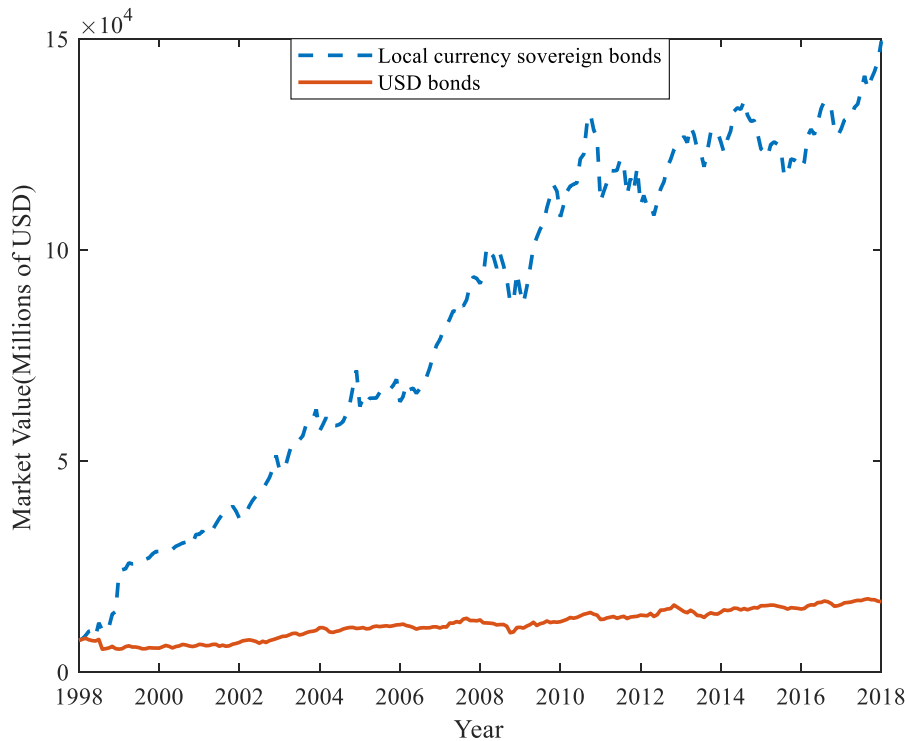


Fig. 4. Sovereign bonds in emerging markets by currency denomination. This figure displays the time series plot of full market value for the local currency sovereign debt and USD sovereign debt. It shows the relative average size of the local currency and USD denominated sovereign bonds of the emerging markets. The data consist of the market value of sovereign bonds in my sample included in the ICE Bank of America Merrill Lynch government bond indexes. Emerging markets (EM) in my sample include countries: Brazil, Chile, China, Colombia, Czech Republic, Egypt, Greece, Hungary, India, Indonesia, Israel, Latvia, Lithuania, Malaysia, Mexico, Morocco, Nigeria, Norway, Pakistan, Peru, Philippines, Poland, Romania, Russia, Slovak Republic, Slovenia, South Africa, South Korea, Sri Lanka, Thailand, Turkey, and Vietnam.

Additionally, I collect monthly bid–ask spreads from Bloomberg.⁹ For each month, I calculate the proportional spread as the ask minus the bid divided by the midpoint price for each bond. The bid–ask spreads of EM sovereign bonds for each country are then calculated as the average of the monthly proportional spreads of all the available sovereign bonds in that month.

2.3. Controls

This paper uses a global asset pricing model as the benchmark (Longstaff, Pan, Pedersen, and Singleton, 2011). I expect that sovereign bond returns are affected by various financial market and macroeconomic variables. To control for these effects, I use the following global control variables. The first set of global control variables include DEF, the default spread (equal to the yield spread between U.S. Moody's BAA and AAA corporate bonds); TERM, the term spread (equal to the yield spread between U.S. ten-year government bonds and three-month Treasury bills); VIX, the CBOE option implied volatility index of the S&P500; the TED spread (the difference between the three-month Libor rate and the U.S. three-month Treasury rate); and the CARRY factor (the difference in excess returns for exchange rates with large and small interest differentials). Many studies use these factors to explain asset returns or to reflect financial conditions (Fama and French, 1993; Baek, Bandoopathyaya, and Du, 2005; Remolona, Scatigma, and Wu, 2008; Hilscher and Nosbusch, 2010; Borri and Verdelhan, 2011; Lustig, Roussanov, and Verdelhan, 2011; Acharya et al., 2013). Data on US short-term Treasury bill rates, US government, and corporate bond yields are obtained from the Federal Reserve Bank of St. Louis database. I obtain the time series data for the foreign exchange factor, denoted as CARRY, from Professor Hanno Lustig's website. Table 2 lists the mean, standard deviation, and correlation matrix of global control variables.

For the unrestricted model, this study uses three types of pricing factors, which are used to explain bond returns. The first factors are the Fama–French six factors: Mkt-RF, SMB, HML, RMW, CMA, and UMD, which are obtained from Professor Kenneth French's website (Fama and French, 2018).¹⁰ SMB, HML, RMW, CMA and UMD are the Size, value-growth, operating prof-

⁹ For local sovereign debt, I collect the bid and ask yields of generic sovereign bonds with different maturities as bid and ask prices are not available in Bloomberg. Then I calculate the bid and ask prices and the proportional spread. For external (USD) sovereign debt, I collect the bid and ask prices for all the sovereign bonds and then calculate the proportional spread directly.

¹⁰ Harvey, Solnik, and Zhou (2002) find that the factors driving bond returns are the same as those driving equity returns.

Table 1

Basic Statistics for Country Data. This table reports the inception year and basic statics of sovereign bond (denominated in local currency) indexes and sovereign bond (denominated in USD) indexes in Panel A and Panel B, respectively. In Panel A, the inception year of Bank of America Merrill Lynch Government total return index (local debt) monthly data ranges from 2005 to 2016. In Panel B, the inception year of Bank of America Merrill Lynch Government total return index (USD debt) monthly data ranges from 1998 to 2013. The end date is October 2018. For each index, I report the mean, St. Dev, and skewness for excess bond returns. Panel A and Panel B display all USD returns, which are measured in excess of the U.S. 1-month Treasury bill returns from Ibbotson Associates. Full market value is in millions of USD. Panel A: Sovereign debt in local currency.

	Inception year	Full market value	Credit rating	Mean (%)	St.Dev (%)	Skewness
Brazil	2005	135441.79	11.28	0.69	5.51	-0.48
Chile	2009	11726.48	17.26	0.16	3.17	-0.32
China	2004	738975.24	16.19	0.35	1.24	0.20
Colombia	2010	56905.63	12.33	0.00	4.65	0.05
Czech Republic	1998	27927.20	16.43	0.51	3.81	-0.23
Egypt	2010	40251.09	6.03	-0.16	7.91	-6.67
Greece	1998	106169.46	11.25	0.35	7.99	-1.19
Hungary	1998	26385.16	13.32	0.44	5.13	-1.06
India	1999	343459.45	10.02	0.38	2.66	-0.03
Indonesia	2004	60135.39	9.86	0.42	5.61	-0.87
Latvia	2014	2602.73	13.86	0.04	2.41	-0.09
Lithuania	2015	2827.36	14.00	-0.02	2.44	-0.27
Malaysia	2006	111921.79	14.96	0.18	2.66	-0.75
Mexico	2001	94204.85	13.62	0.17	3.99	-0.68
Morocco	2010	27265.14	11.11	0.20	1.90	-0.69
Nigeria	2011	17451.13	7.89	0.05	5.92	-3.19
Pakistan	2016	18092.80	5.40	-0.62	2.35	-1.87
Peru	2011	16314.90	13.34	0.24	3.18	-0.09
Philippines	2004	47331.16	10.00	0.61	3.11	-0.42
Poland	1998	68634.75	14.94	0.53	4.56	-0.58
Romania	2011	18679.84	11.00	0.15	2.91	-0.15
Russia	2005	50827.28	12.57	-0.01	6.38	-1.11
Slovakia	2003	23801.31	15.40	0.45	3.22	-0.42
Slovenia	2007	16503.50	15.27	0.23	3.90	-0.66
South Africa	1998	60577.37	14.02	0.26	6.29	-0.47
South Korea	2000	236601.95	16.00	0.34	3.57	0.24
Sri Lanka	2015	17877.93	7.00	-0.11	2.71	-1.10
Thailand	2002	66696.18	14.04	0.38	2.47	-0.02
Turkey	2004	53122.72	9.95	-0.16	6.86	-1.85
Vietnam	2014	25066.69	7.93	0.36	1.38	-0.13

Panel B: Sovereign debt in USD

	Inception year	Full market value	Credit rating	Mean (%)	St.Dev (%)	Skewness
Brazil	1998	39235.03	9.29	0.59	4.99	-1.50
Chile	1999	2943.40	14.92	0.31	1.61	-0.12
China	1998	2980.38	14.51	0.26	1.10	0.10
Colombia	1998	12736.81	10.38	0.56	3.22	-1.12
Egypt	2001	3158.66	8.41	0.45	2.88	-1.20
Greece	1998	1698.15	13.01	-0.38	5.18	-5.21
Hungary	1998	5805.31	12.47	0.37	2.77	-3.11
Indonesia	1999	19944.95	8.03	0.57	3.53	-1.57
Latvia	2011	1886.05	12.82	0.36	1.41	-0.03
Lithuania	2009	6236.05	12.86	0.52	1.84	-0.51
Malaysia	1999	3033.18	13.41	0.35	1.50	-0.45
Mexico	1998	35935.35	11.60	0.45	2.59	-1.52
Morocco	1998	1699.56	9.74	0.20	3.16	-3.27
Nigeria	1998	2663.32	7.64	0.62	5.05	-2.28
Pakistan	2000	1989.84	5.00	0.75	4.34	-1.35
Peru	1998	8044.88	10.14	0.59	4.19	-1.80
Philippines	1998	20211.66	9.51	0.57	2.82	-2.55
Poland	1998	8409.76	13.50	0.37	1.77	-1.21
Romania	2012	5725.78	11.00	0.53	2.02	-0.18
Russia	1998	33997.74	9.60	0.56	9.18	-8.87
Slovakia	2012	1182.71	13.12	0.52	1.34	0.56
Slovenia	2013	6795.55	12.60	0.42	1.64	-0.50
South Africa	1998	6790.96	11.93	0.44	2.57	-1.54
Sri Lanka	2007	4390.29	6.81	0.53	5.68	-5.49
Thailand	1998	1003.09	11.58	0.35	1.75	-1.25
Turkey	1998	28986.52	8.13	0.53	3.79	-1.32
Vietnam	2005	1465.54	7.68	0.45	3.19	-4.35

Table 2

Summary statistics for sentiment and controls, This table shows the summary statistics and correlation matrix of sentiment and control variables. This table lists monthly average of variables from January 1998 to September 2015. Mkt-RF-UMD are the Fama-French 6-factor constructed on the global market. TED-TERM are liquidity spread, default spread, and term spread. VIX is the implied volatility of the S&P500 index. CARRY is the foreign exchange factor and is from Lustig et al. (2011). CP2005 is the predictable component of bond returns of Cochrane-Piazzesi (2005) and LN2009 is the dynamic factor from Ludvigson and Ng (2009).

Panel A: Mean and Standard Deviation																
	S^{BW}	$S^{BW\perp}$	$S_{lag}^{BW\perp}$	Mkt-RF	SMB	HML	RMW	CMA	UMD	CP2005	LN2009	VIX	TED	TERM	DEF	CARRY
Mean (%)	0.02	0.16	0.16	0.45	0.20	0.37	0.34	0.31	0.38	1.56	1.66	20.91	0.04	0.15	0.22	0.69
St.Dev	0.67	0.68	0.66	4.55	1.98	2.62	1.60	2.08	5.29	1.07	1.62	8.03	0.03	0.10	0.06	2.39
Panel B: Correlation matrix																
	S^{BW}	$S^{BW\perp}$	$S_{lag}^{BW\perp}$	Mkt-RF	SMB	HML	RMW	CMA	UMD	CP2005	LN2009	VIX	TED	TERM	DEF	CARRY
S^{BW}	1															
$S^{BW\perp}$	0.97	1														
$S_{lag}^{BW\perp}$	0.95	0.96	1													
Mkt-RF	-0.21	-0.20	-0.20	1												
SMB	0.00	-0.01	-0.03	-0.01	1											
HML	0.32	0.34	0.28	-0.18	-0.01	1										
RMW	0.23	0.24	0.22	-0.50	-0.23	0.34	1									
CMA	0.33	0.33	0.29	-0.47	-0.04	0.75	0.34	1								
UMD	0.12	0.12	0.11	-0.31	0.10	-0.14	0.22	0.08	1							
CP2005	-0.04	-0.08	-0.08	-0.07	0.05	0.05	0.10	0.08	0.01	1						
LN2009	0.27	0.20	0.24	-0.06	0.04	0.07	0.05	0.08	-0.16	0.05	1					
VIX	0.02	-0.09	-0.05	-0.43	-0.06	-0.09	0.18	0.13	-0.02	0.22	0.45	1				
TED	0.07	0.03	0.05	-0.28	-0.24	-0.09	0.02	0.09	0.05	-0.08	0.24	0.44	1			
TERM	-0.45	-0.48	-0.44	0.03	0.16	-0.08	-0.03	-0.03	0.08	0.22	0.18	0.18	-0.28	1		
DEF	-0.14	-0.22	-0.18	-0.19	0.00	-0.03	0.08	0.10	0.14	0.13	0.61	0.73	0.25	0.48	1	
CARRY	0.01	0.00	0.01	0.45	0.08	-0.06	-0.20	-0.22	-0.16	0.09	-0.04	-0.19	-0.20	-0.08	-0.18	1

itability, investment, and momentum factors respectively.¹¹ The second set of factors are based on forward rates (Fama and Bliss, 1987; Cochrane and Piazzesi, 2005). Cochrane and Piazzesi find that a single linear combination of one- to five-year forward rates explains between 31% and 37% of next year's excess bond returns. I use the Fama-Bliss dataset available from the Center for Research in Security Prices (CRSP) from October 1995 to October 2018. It contains observations on one to five-year zero-coupon US Treasury bond prices. Following Cochrane and Piazzesi (2005), I estimate this linear combination of forward rates for the US market in the unrestricted model and denote this factor as CP2005_t. Moreover, I consider the role of macroeconomic variables with statistical power to explain variation on the bond returns and use a macroeconomic factor as the third type of pricing factor (Ludvigson and Ng, 2009). Ludvigson and Ng summarize the information on 132 monthly economic series into a set of dynamic factors. I obtain the macro factors available at Professor Sydney Ludvigson's web paper (<https://www.sydneyludvigson.com/data-and-appendixes>), and estimate this macro-economic factor (denoted as LN2009_t).

In the analysis, this paper also controls for a number of country-specific variables including the GDP growth rate, inflation rate, foreign exchange rate return, sovereign credit rating, external debt, and the lagged EM sovereign bond excess returns (Butler and Fauver, 2006; Greenwood and Vayanos, 2014; Hilscher and Nosbusch, 2010; Jeanneret and Souissi, 2016). To proxy for a country's political stability, I follow Csoneto and Ivashchenko (2013) and use a political risk index. The political risk index from International Country Risk Guide (ICRG) assesses countries' political stability. This political risk index ranges from zero (highest risk) to 100 (lowest risk). I obtain country variables (such as GDP, inflation rate, and external debt) from World Development Indicators (WDI) and the International Monetary Fund (IMF). Tables 3 lists means for selected variables by country and Table 4 lists the descriptive statistics for the full sample.

3. Empirical results

In this section, I follow Stambaugh, Yu, and Yuan (2012), Baker and Wurgler (2012), and Kim, Kim, and Seo (2017) and use regression models to investigate how the investor sentiment index is related to future EM sovereign bond returns.¹² I investigate sentiment effects on the sovereign debt issued in local currency in Section 3.1 and in USD in Section 3.2. I first examine how investor sentiment predicts bond returns under well-known global economic variables and pricing factors. Second, I control for additional local macro variables in the testing model. My interest is to investigate whether there is a significant relationship between investor sentiment and EM sovereign bond returns.

¹¹ Mkt-RF, SMB, HML, RMW, and CMA I use in this paper are the Fama-French global five factors. Fama and French (2018) further add the momentum factor, UMD into their 5-factor model to form their 6-factor model.

¹² My regression models are not real time predictive regressions. My regression models include additional controls to analyze EM sovereign bond returns in Section 3.3. In unreported regressions, I control for lagged global variables and lagged local controls. The estimated coefficients on sentiment are qualitatively similar to the results in Table 5-8.

Table 3

Descriptive statistics of local macro variables from 1998 to year 2016. This table reports the means of GDP growth rate (%), domestic investment, exchange rate return, inflation rate, political risk index, total reserves (includes gold), external debt, and foreign investment for each country. GDP growth rate is the annual percentage growth rate of GDP. Domestic investment, total reserves (includes gold), external debt, and foreign investment are the ratios to GDP.

Country	GDP growth (%)	Domestic investment	Exchange rate return	Inflation rate	Political risk index	Total reserves	External debt	Foreign investment
Brazil	2.34	18.99	0.68	6.54	65.88	11.07	25.36	3.35
Chile	3.82	22.94	0.24	3.39	76.90	16.28	44.66	7.41
China	9.25	42.22	-0.08	1.89	64.07	32.30	12.66	3.42
Colombia	3.44	20.38	0.44	6.24	56.50	10.92	29.21	3.57
Czech Republic	2.56	28.68	-0.07	2.64	77.92	23.64	43.35	5.42
Egypt	4.44	17.98	0.89	8.13	58.63	15.02	24.52	2.72
Greece	0.51	20.28	0.10	2.30	73.93	4.06	173.15	0.72
Hungary	2.34	23.96	0.24	5.35	78.01	23.91	122.37	11.45
India	7.13	34.40	0.26	6.90	60.09	15.94	19.72	1.53
Indonesia	4.09	28.26	0.67	10.57	53.79	14.24	52.45	0.93
Latvia	3.97	27.64	0.09	3.79	72.81	17.36	125.40	3.73
Lithuania	4.08	21.43	-0.05	2.40	73.48	15.12	65.16	3.32
Malaysia	4.47	23.84	0.13	2.42	72.41	40.87	65.46	3.38
Mexico	2.39	22.62	0.46	5.77	69.36	9.90	24.53	2.71
Morocco	4.26	31.55	0.04	1.64	69.20	22.28	36.83	2.10
Nigeria	5.87	23.99	0.69	11.42	44.43	11.66	18.58	1.77
Pakistan	4.12	16.70	0.39	7.72	47.73	7.57	35.23	1.26
Peru	4.62	21.48	0.10	3.08	62.89	24.38	38.80	3.90
Philippines	4.80	20.53	0.12	4.28	64.01	23.60	47.51	1.55
Poland	3.76	21.53	0.15	3.36	77.80	16.99	53.86	3.55
Romania	3.31	24.70	0.80	15.09	68.52	19.75	48.95	3.81
Russia	3.62	21.25	1.21	16.35	60.95	21.94	39.91	2.25
Slovakia	3.72	26.51	-0.04	4.26	76.61	12.43	63.86	4.25
Slovenia	2.38	25.18	0.17	3.72	77.19	11.17	91.46	2.00
South Africa	2.82	19.13	0.57	5.48	66.90	9.85	27.10	1.55
South Korea	4.06	31.04	-0.08	2.78	76.70	23.10	28.07	1.10
Sri Lanka	5.39	28.54	0.40	8.40	56.19	10.19	51.25	1.24
Thailand	3.46	24.45	-0.10	2.43	62.37	35.41	40.32	3.18
Turkey	4.52	26.07	1.37	22.79	58.13	11.08	40.63	1.49
Vietnam	6.27	31.77	0.23	6.86	65.39	15.42	41.24	5.42

Table 4

Descriptive Statistics for all EMs. This table lists means, standard deviations, 25th percentiles, 50th percentiles, and 75th percentiles for EM bond bid-ask spreads, credit ratings, and other macro variables of foreign exchange rate return, GDP growth rate, inflation rate, and political risk index. Credit rating: A numerical translation of EM bond rating: 20 = AAA and 0 = D (see Appendix). Political risk index is the International Country Risk Guide (ICRG) political risk index of countries' political stability. This political risk index ranges from zero (highest risk) to 100 (lowest risk). GDP growth rate is the annual percentage growth rate of GDP.

Variable	Mean	Std. Dev.	25th percentile	50th percentile	75th percentile
Bid-ask spread of local debt (%)	0.13	0.30	0.04	0.07	0.14
Bid-ask spread of external debt (%)	0.15	0.14	0.07	0.12	0.17
Credit rating of local debt	12.83	3.22	11	13	15
Credit rating of USD external debt	10.71	3.09	8	11	13
Foreign exchange return (%)	0.31	3.87	-1.08	0.01	1.37
Political risk index	66.25	9.82	60.25	67.08	74.58
GDP growth rate (%)	4.06	3.53	2.59	4.36	6.19
Inflation rate (%)	6.18	8.57	2.26	4.11	7.42

3.1. EM local sovereign debt

This subsection empirically assesses the predictive performance of market sentiment on EM local currency sovereign bond returns using regression models (1) and (2):

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S^{BW\perp}_{t-1} + \beta_2 Global_t + \varepsilon_{i,t}, \quad (1)$$

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S^{BW\perp}_{t-1} + \beta_2 Global_t + \beta_3 \Phi_t + \varepsilon_{i,t}, \quad (2)$$

where $R_{i,t}^{LOC}$ are the returns of EM sovereign debt denominated in local currency; $Global_t$ includes the implied volatility of the S&P500 index (VIX), the liquidity spread (TED), term spread (TERM), default spread (DEF), and carry trade factor (CARRY); Φ_t includes the pricing factors from [Cochrane and Piazzesi \(2005\)](#)'s tent-shaped "single return-forecasting factor", [Ludvigson](#)

and Ng (2009)'s dynamic macro-economic factor, as well as Fama and French (2018) six factors (Mkt-RF, SMB, HML, RMW, CMA, and UMD).¹³

I hypothesize that there is a relationship between the EM sovereign bond returns and investor sentiment. In Panel A of Table 5, I use the excess returns for EM local sovereign debt valued in US dollars as the dependent variable and control for contemporaneous global factors to investigate whether sentiment predicts bond returns beyond the information in the global factors in column (1). I find that the estimated coefficients on sentiment in Panel A and Panel B of Table 5 are negative and significant.¹⁴ Quantitatively, the slope coefficient of -1.152 in column (1) indicates that a one standard deviation increase in sentiment is associated with a monthly 76 basis point decrease in the subsequent EM sovereign bond returns denominated in local currency (multiplying the standard deviation of lagged sentiment, 0.66, by the estimated coefficient of -1.152). The estimated coefficient on sentiment is economically and statistically significant. I next control for the forward rates-based factor (CP2005), a common macroeconomic factor (LN2009), and the Fama-French six-factors respectively in column (2)–(4).¹⁵ After adjusting for benchmark exposures, the estimated coefficients on sentiment are negative and significant at the 1 percent level with the exception of column (4) where the coefficient on sentiment is significant at the 5 percent level. After adjusting for the Fama-French six factors in column (4), the estimated coefficient on investor sentiment is -0.664 , which is about half the magnitude of the coefficient in column (1) but is nevertheless significant (t -statistic: -2.3). Economically, the coefficient in column (4) implies that a one standard deviation increase in sentiment is associated with a 43.8 basis point decrease in EM sovereign bond returns (multiplying the standard deviation of lagged sentiment, 0.66, by the estimated coefficient of -0.664). Compared to other pricing factors and control variables, the market factor (Mkt-RF) has the largest effect on EM bond returns: a one standard deviation increase in the market factor is associated with 225 basis point increase in EM sovereign bond returns (the standard deviation of Mkt-RF is 4.55). The VIX has the smallest significant effect on EM bond return: a one standard deviation increase in the VIX is associated with 33.7 basis point increase in EM sovereign bond returns (the standard deviation of VIX is 20.91). The monthly average excess returns for the whole sample of EM sovereign debt issued in local currency is 39 basis points.

In Panel B of Table 5, I use the excess returns of EM local sovereign currency debt valued in the local currency as the dependent variable. As above, the estimated coefficient on sentiment is negative and significant at the 1 percent level. In column (1), a one standard deviation increase in sentiment is associated with a 44.3 basis point decrease in the subsequent EM sovereign bond returns, a slightly smaller sentiment impact when local sovereign debt is valued in the local currency than in US dollars. The Fama-French four factors, SMB, HML, RMW, and CMA, lose their significance when returns are valued in the local currency. I also note that the R^2 s of the regressions in Panel B are much smaller than those in Panel A, and this lower R^2 is consistent with the local currency returns containing additional noise that is not well captured by the global factors.

Taken together, sentiment contains information that is not fully captured by the common factors and helps predict sovereign bond returns of EM local debt. When the sentiment index is low, investors are more pessimistic about risky assets and react more to negative information (Mian and Sankaraguruswamy, 2012), thus there is a flight-to-safety and EM sovereign bonds are not perceived as safe assets (Erb, Harvey, and Viskanta, 1999; Bekaert and Harvey, 2017).

3.2. EM external sovereign debt

In this subsection I empirically assess the predictive performance of market sentiment on EM USD sovereign bond returns in the regression model (3):

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S_{t-1}^{BW} + \beta_2 Global_t + \beta_3 \Phi_t + \varepsilon_{i,t}, \quad (3)$$

where $R_{i,t}^{USD}$ are the returns on EM sovereign debt denominated in USD.

Consistent with the results for EM local sovereign debt, sentiment is negatively related to the future bond returns of external (USD) debt in Table 6. The slope coefficients on sentiment are all significant at the 1 percent level. According to the estimated results in column (4), a one standard deviation increase in sentiment is associated with a 50.2 basis point reduction in the subsequent bond return of external sovereign debt, thus there is a slightly stronger impact on sovereign debt denominated in USD than on sovereign debt denominated in local currency. I note that the t -statistics of coefficients on sentiment are greater in Table 6 than those in Table 5. This stronger sentiment effect on bonds denominated in USD than LOC is consistent with US investor sentiment being more closely related to sovereign bonds denominated in USD (Longstaff et al., 2011; Du and Schreger, 2016).

¹³ In unreported regressions, I also control for changes in the VIX and volatility risk premium, the difference between the VIX and a measure of realized volatility for the S&P100 index (see Longstaff, Pan, Pedersen, and Singleton, 2011). The estimated coefficients on investor sentiment are qualitatively similar to the results in Table 5–8.

¹⁴ Petersen (2009) argues that both OLS and Fama-MacBeth standard error estimates are biased down significantly in the presence of a fixed firm effect, and suggests to cluster standard errors in both dimensions. In unreported regressions, I cluster standard errors by both country and time. For sentiment effects on sovereign bond returns, the results are qualitatively similar to Table 5–Table 8.

¹⁵ In unreported regressions, I also control for CP2005, LN2009, and the Fama-French six-factor simultaneously. The estimated coefficient on investor sentiment is qualitatively similar to the results in Table 5.

Table 5

Sentiment and future sovereign returns of local currency debt with global controls and factors The table reports the estimated results in the regressions:

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S_{lag}^{BW\perp} + \beta_2 Global_t + \varepsilon_{i,t}, \text{column(1)}$$

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S_{lag}^{BW\perp} + \beta_2 Global_t + aCP2005_t + \varepsilon_{i,t}, \text{column(2)}$$

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S_{lag}^{BW\perp} + \beta_2 Global_t + bLN2009_t + \varepsilon_{i,t}, \text{column(3)}$$

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S_{lag}^{BW\perp} + \beta_2 Global_t + cFF6_t + \varepsilon_{i,t}, \text{column(4)},$$

where $R_{i,t}^{LOC}$ is excess return of sovereign bond denominated in local currency in month t valued either in USD (panel a) or LOC (panel b), $S_{lag}^{BW\perp}$ is the level of the orthogonal investor-sentiment index of Baker and Wurgler (2006), $Global_t$ represents global controls of VIX, TED, TERM, DEF and CARRY, $CP2005_t$ is the predictable component of bond returns of Cochrane-Piazzesi (2005), $LN2009_t$ is the dynamic factor from Ludvigson and Ng (2009), and $FF6_t$ represents the Fama-French six factors of Mkt-RF, SMB, HML, RMW, CMA, and UMD (Fama and French, 2018). The dependent variable in Panel A is USD return of sovereign debt issued in local currency in excess of the U.S. 1-month Treasury bill returns from Ibbotson Associates, while that in Panel B is LOC return of debt issued in local currency in excess of local short-term treasury yield. t values are calculated using standard errors clustered by country. Country fixed effects and year fixed effects are included. ***, **, and * denote t -statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

Panel A: Sentiment effects on sovereign bond USD returns of local currency issued debt

	(1)	(2)	(3)	(4)
$S_{lag}^{BW\perp}$	-1.152*** (-4.29)	-1.095*** (-4.13)	-1.115*** (-4.17)	-0.664** (-2.30)
CP2005		-0.368*** (-3.55)		
LN2009			0.408*** (3.93)	
Mkt-RF				0.495*** (9.61)
SMB				0.235*** (4.18)
HML				0.206*** (2.86)
RMW				0.248*** (3.82)
CMA				-0.036 (-0.58)
UMD				-0.005 (-0.25)
VIX	-0.241*** (-8.18)	-0.250*** (-8.30)	-0.220*** (-8.17)	0.042** (2.55)
TED	-0.739 (-1.58)	-0.724 (-1.55)	-1.077** (-2.44)	-0.160 (-0.36)
TERM	-0.326* (-1.96)	-0.336* (-2.04)	-0.145 (-0.82)	-0.243 (-1.37)
DEF	2.076*** (6.28)	2.250*** (6.56)	1.805*** (5.31)	0.094 (0.33)
CARRY	0.514*** (6.01)	0.514*** (6.02)	0.516*** (6.03)	0.233*** (2.81)
Constant	4.234*** (3.64)	4.857*** (3.99)	3.899*** (3.45)	0.200 (0.20)
Observations	3200	3200	3200	3200
R^2	0.163	0.166	0.169	0.260
Adjusted R^2	0.150	0.153	0.155	0.246

Panel B: Sentiment effects on sovereign bond LOC returns of LOC issued debt

	(1)	(2)	(3)	(4)
$S_{lag}^{BW\perp}$	-0.671*** (-4.32)	-0.645*** (-4.28)	-0.649*** (-4.19)	-0.619*** (-3.49)
CP2005		-0.158 (-1.61)		
LN2009			0.211*** (3.41)	
Mkt-RF				0.112*** (4.99)
SMB				0.0137 (0.42)
HML				0.029 (0.48)
RMW				0.086 (1.44)
CMA				0.057 (1.00)
UMD				-0.023** (-2.29)

(continued on next page)

Table 5 (continued)

Panel B: Sentiment effects on sovereign bond LOC returns of LOC issued debt				
	(1)	(2)	(3)	(4)
VIX	−0.037** (−2.21)	−0.041** (−2.24)	−0.026* (−1.73)	0.024** (2.35)
TED	−0.293 (−1.15)	−0.284 (−1.11)	−0.476* (−1.97)	−0.286 (−1.23)
TERM	−0.307*** (−3.47)	−0.313*** (−3.61)	−0.212** (−2.10)	−0.263** (−2.61)
DEF	0.552** (2.47)	0.625** (2.62)	0.417* (1.85)	0.126 (0.65)
CARRY	0.129*** (3.91)	0.129*** (3.92)	0.129*** (3.93)	0.066*** (3.07)
Constant	0.741 (1.23)	1.009 (1.42)	0.562 (0.96)	−0.064 (−0.12)
Observations	3061	3061	3061	3061
R ²	0.042	0.044	0.047	0.061
Adjusted R ²	0.026	0.028	0.031	0.044

Table 6

Sentiment and future sovereign returns of USD debt with global controls and factors, The table reports the estimated results in the regressions,

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S_{log}^{BW\perp} + \beta_2 Global_t + \varepsilon_{i,t}, \text{ column(1) ,}$$

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S_{log}^{BW\perp} + \beta_2 Global_t + \alpha CP2005_t + \varepsilon_{i,t}, \text{ column(2) ,}$$

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S_{log}^{BW\perp} + \beta_2 Global_t + \alpha LN2009_t + \varepsilon_{i,t}, \text{ column(3) ,}$$

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S_{log}^{BW\perp} + \beta_2 Global_t + \alpha FFG_t + \varepsilon_{i,t}, \text{ column(4) ,}$$

where $R_{i,t}^{USD}$ is excess USD return of sovereign bond issued in USD in month t , $S_{log}^{BW\perp}$ is the level of the orthogonal investor-sentiment index of Baker and Wurgler (2006), $Global_t$ represents global controls of VIX, TED, TERM, DEF and CARRY, $CP2005_t$ is the predictable component of bond returns of Cochrane-Piazzesi (2005), $LN2009_t$ is the dynamic factor from Ludvigson and Ng (2009), and FFG_t represents Fama-French six factors of Mkt-RF, SMB, HML, RMW, CMA, and UMD (Fama and French, 2018). t values are calculated using standard errors clustered by country. Country fixed effects and year fixed effects are included. ***, **, and * denote t -statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	(1)	(2)	(3)	(4)
$S_{log}^{BW\perp}$	−1.202*** (−6.04)	−1.176*** (−6.07)	−1.221*** (−6.11)	−0.761*** (−3.99)
CP2005		−0.270** (−2.59)		
LN2009			0.700*** (7.46)	
Mkt-RF				0.246*** (7.85)
SMB				0.061* (2.00)
HML				0.118** (2.58)
RMW				0.062 (1.28)
CMA				−0.224** (−2.37)
UMD				0.043*** (3.60)
VIX	−0.275*** (−5.65)	−0.280*** (−5.60)	−0.237*** (−5.29)	−0.118*** (−3.52)
TED	−0.478 (−1.66)	−0.486 (−1.69)	−0.821*** (−2.91)	−0.244 (−0.96)
TERM	−0.741*** (−6.97)	−0.767*** (−7.19)	−0.471*** (−4.09)	−0.748*** (−7.00)
DEF	2.839*** (6.46)	2.953*** (6.30)	2.182*** (5.74)	1.787*** (5.22)
CARRY	0.268*** (6.63)	0.266*** (6.66)	0.265*** (6.60)	0.135*** (5.88)
Constant	1.112 (1.68)	1.566*** (2.82)	0.710 (1.02)	−1.312 (−1.48)
Observations	4246	4246	4246	4246
R ²	0.144	0.146	0.166	0.183
Adjusted R ²	0.134	0.135	0.156	0.172

3.3. Controlling for additional local macro variables

Local, rather than US, macroeconomic conditions may capture some of the effects of sentiment and drive EM sovereign bond returns. To assess this potential explanation, I control for an additional set of local macro variables in model (4): the exchange rate return, sovereign bond credit rating, political risk, the GDP growth rate, the inflation rate, and the lagged EM sovereign bond return,

$$R_{i,t} = \alpha_i + \beta_1 S^{BW\perp}_{t-1} + \beta_2 Global_t + \beta_3 \Phi_t + \beta_4 Local_{i,t} + \varepsilon_{i,t}, \quad (4)$$

where $R_{i,t}$ is the returns on EM sovereign debt denominated either in local currency or in USD; $Local_t$ includes the exchange rate return, sovereign bond credit rating, political risk index, the GDP growth rate, the inflation rate, and the lagged bond return.¹⁶

Using model (4), I test whether the ability of investor sentiment to predict benchmark-adjusted returns is robust to controlling for local macro-related fluctuations. Table 7 reports the results from regressing excess returns of EM local sovereign debt on the lagged sentiment index, the contemporaneous returns of other commonly used factors, and five macro-related variables. The effects of investor sentiment remain similar after I consider the additional local macro-related variables. The estimated coefficients on sentiment are all significant at the 5 percent level. In column (4) of Panel A, which is based on returns adjusted for exposures to the Fama-French six-factor and global controls, a one standard deviation increase in sentiment is associated with a 25.3 basis point decrease in EM local sovereign bond returns. Compared to Table 5, the magnitudes of the sentiment effect are smaller after I consider local macro variables.

Table 8 repeats the analyses in Table 7 for excess returns of EM sovereign bonds denominated in USD.¹⁷ The coefficients and their t -statistics on sentiment are close to those in Table 6 which do not include the additional local macro variables. As indicated in column (4) of Table 8, a one standard deviation increase in sentiment is associated with a 52.4 basis point reduction in future sovereign returns. I note that most of the Fama-French factors, the CP factor, and global control variables are not significant in explaining returns of local debt in Table 7, but remain significant for USD debt. When I control for local macro variables, the t -statistics of pricing factors and global controls shrink in models of EM local sovereign bond returns than external (USD) sovereign bond returns.

To conclude this subsection, I find that the estimated coefficients on sentiment remain negative and significant when I control for additional local macro variables. The results are consistent for both EM local debt and external (USD) debt. When sentiment is high, subsequent EM sovereign bond returns tend to be low due to a later correction. Overall, the negative impact of sentiment is economically and statistically significant.

3.4. Sovereign debt in developed countries

In this subsection, I examine the predictive ability of investor sentiment for sovereign bonds in developed markets. The analysis of developed markets in this paper includes a sample of 26 developed countries (see Section 2.2). Previous studies mainly study US bonds markets or one specific European country. In Table 9, the estimated coefficients of sentiment are positive and statistically significant at the 5 percent level. Economically, one standard deviation drop in sentiment is associated with a 24.4 basis point decrease in DM sovereign bond returns according to the results in column (2). The results are consistent with previous studies for US bond markets (Laborda and Olmo, 2014; Baker and Wurgler, 2012).¹⁸ When investor sentiment is low, subsequent sovereign bond returns for developed countries are lower. On the other hand, subsequent EM sovereign bond returns are relatively high when sentiment is low; investors' flight-to-safety drives up the prices of sovereign bonds in developed markets and pulls down the prices of EM sovereign bonds.

Taken together, sentiment effects on sovereign bonds returns vary among markets. While sentiment has positive and significant effects on future sovereign bond returns among developed countries, sentiment impact on future sovereign bond returns among EMs is negative and significant. The results are consistent with investors treating EM sovereign bonds as risky assets.

¹⁶ One may argue that using FX (the exchange rate return) as a control variable in Table 7 may cause an endogeneity issue. However, the estimated coefficients on sentiments remain negative and significant with or without FX in the regression. The estimated coefficients on sentiments even have a larger t -statistic without FX in the models of Table 7.

¹⁷ I repeat the baseline regression models in Table 7 and Table 8 for each sample country. The loadings on sentiment are generally negative for EM sovereign bonds issued in LOC and are all negative for bonds issued in USD. The results for individual country tests need to be interpreted with caution due to the smaller sample size. In unreported regressions, I create credit rating dummies for each rating category and run the regressions of model (4) with credit rating dummies. The results of sentiment are qualitatively similar to the results in Table 7 and Table 8.

¹⁸ Baker and Wurgler (2012) find that a one-standard deviation increase in sentiment is associated with 16 basis point higher excess returns on intermediated-term bonds and 26 basis point higher excess returns on long-term bonds. My estimated economic magnitude of 24.4 basis point is close to their estimates. I test the role of sentiment on sovereign bonds from developed countries for the post-crisis period. The sentiment results are qualitatively similar to the results in Table 9. I also repeat the post-crisis analysis for those countries whose credit ratings were downgraded by the large three agencies (Cyprus, Ireland, Spain, and Portugal) using robust regressions accounting for outliers (Yohai, 1987; Maronna and Yohai, 2000; Adams, Hayunga, Mansi, Reeb, and Verardi, 2019). The sentiment effects are not significant for this smaller sample.

Table 7

Sentiment and future sovereign returns of local currency debt, controlling for additional macro variables The table reports the estimated results in the regressions

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S_{lag}^{BW\perp} + \beta_2 Global_t + \beta_3 Country_{i,t} + e_{i,t}, \text{column(1)}$$

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S_{lag}^{BW\perp} + \beta_2 Global_t + \alpha CP2005_t + \beta_3 Country_{i,t} + e_{i,t}, \text{column(2)}$$

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S_{lag}^{BW\perp} + \beta_2 Global_t + \beta LN2009_t + \beta_3 Country_{i,t} + e_{i,t}, \text{column(3)}$$

$$R_{i,t}^{LOC} = \alpha_i + \beta_1 S_{lag}^{BW\perp} + \beta_2 Global_t + cFF6_t + \beta_3 Country_{i,t} + e_{i,t}, \text{column(4)},$$

where $S_{lag}^{BW\perp}$ is the level of the orthogonal investor-sentiment index of Baker and Wurgler (2006), $Global_t$, $CP2005_t$, $LN2009_t$, and $FF6_t$, are described in Table 5. $Country_{i,t}$ includes FX (exchange rate return), credit rating, political risk index, the GDP growth rate, the inflation rate, and R_{lag} (the lagged bond return). The dependent variable $R_{i,t}^{LOC}$ in Panel A is USD return of sovereign debt issued in local currency in excess of the U.S. 1-month Treasury bill returns from Ibbotson Associates, while that in Panel B is LOC return of debt issued in local currency in excess of local short-term treasury yield. Pakistan and Sri Lanka are not in the sample as limited data availability. t values are calculated using standard errors clustered by country. Country fixed effects and year fixed effects are included. ***, **, and * denote t -statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

Panel A: Sentiment effects on sovereign bond USD returns of debt issued in local currency

	(1)	(2)	(3)	(4)
$S_{lag}^{BW\perp}$	−0.385*** (−3.03)	−0.373*** (−2.90)	−0.379*** (−3.03)	−0.384** (−2.08)
CP2005		−0.094 (−1.10)		
LN2009			0.139** (2.56)	
Mkt-RF				0.055*** (3.17)
SMB				−0.018 (−0.57)
HML				−0.024 (−0.44)
RMW				0.031 (0.53)
CMA				0.094* (1.82)
UMD				−0.022** (−2.74)
FX	−1.188*** (−35.41)	−1.187*** (−35.86)	−1.185*** (−35.06)	−1.171*** (−38.47)
Credit rating/100	−0.011 (−0.00)	0.087 (0.02)	0.045 (0.01)	0.108 (0.02)
Political risk index/100	0.005 (0.25)	0.005 (0.23)	0.005 (0.24)	0.004 (0.21)
GDP growth	−0.022 (−0.54)	−0.022 (−0.54)	−0.022 (−0.54)	−0.022 (−0.53)
Inflation	−0.018 (−0.70)	−0.018 (−0.70)	−0.018 (−0.69)	−0.017 (−0.67)
R_{lag}	0.055** (2.73)	0.054*** (2.79)	0.053** (2.66)	0.057** (2.69)
VIX	0.020* (1.78)	0.017 (1.36)	0.026** (2.53)	0.043*** (3.63)
TED	−0.172 (−0.73)	−0.169 (−0.72)	−0.289 (−1.31)	−0.284 (−1.36)
TERM	−0.200** (−2.16)	−0.204** (−2.26)	−0.140 (−1.36)	−0.165 (−1.56)
DEF	0.370* (1.92)	0.416* (2.03)	0.281 (1.43)	0.214 (1.18)
CARRY	0.073** (2.47)	0.073** (2.48)	0.074** (2.48)	0.049* (1.99)
Constant	1.029 (0.78)	1.215 (0.87)	0.942 (0.71)	0.805 (0.63)
Observations	3172	3172	3172	3172
R^2	0.734	0.734	0.735	0.736
Adjusted R^2	0.729	0.729	0.730	0.731

Panel B: Sentiment effects on sovereign bond LOC returns of debt issued in local currency

	(1)	(2)	(3)	(4)
$S_{lag}^{BW\perp}$	−0.444*** (−3.30)	−0.430*** (−3.17)	−0.432*** (−3.24)	−0.474** (−2.32)
CP2005		−0.095 (−1.22)		
LN2009			0.161** (2.77)	

Table 7 (continued)

Panel B: Sentiment effects on sovereign bond LOC returns of debt issued in local currency				
	(1)	(2)	(3)	(4)
Mkt–RF				0.044** (2.53)
SMB				–0.034 (–1.02)
HML				–0.012 (–0.23)
RMW				0.040 (0.65)
CMA				0.084 (1.51)
UMD				–0.023*** (–2.94)
FX	–0.200*** (–5.66)	–0.198*** (–5.69)	–0.197*** (–5.56)	–0.186*** (–5.79)
Credit rating/100	0.310 (0.07)	0.412 (0.09)	0.404 (0.09)	0.426 (0.09)
Political risk index	0.014 (0.61)	0.013 (0.58)	0.013 (0.59)	0.013 (0.58)
GDP growth/100	–0.542 (–0.15)	–0.539 (–0.15)	–0.554 (–0.15)	–0.513 (–0.14)
Inflation	–0.048* (–1.98)	–0.048* (–1.97)	–0.047* (–1.96)	–0.046* (–1.96)
R_{lag}	0.132*** (4.02)	0.130*** (4.14)	0.132*** (4.05)	0.131*** (3.85)
VIX	0.006 (0.51)	0.004 (0.26)	0.014 (1.24)	0.024** (2.21)
TED	–0.168 (–0.71)	–0.163 (–0.69)	–0.308 (–1.42)	–0.301 (–1.44)
TERM	–0.248*** (–2.85)	–0.252*** (–2.97)	–0.175* (–1.77)	–0.213** (–2.11)
DEF	0.273 (1.39)	0.318 (1.54)	0.174 (0.87)	0.159 (0.90)
CARRY	0.063* (1.98)	0.063* (1.99)	0.064* (2.02)	0.044 (1.70)
Constant	–0.341 (–0.22)	–0.152 (–0.09)	–0.465 (–0.30)	–0.506 (–0.33)
Observations	3032	3032	3032	3032
R^2	0.112	0.113	0.115	0.117
Adjusted R^2	0.095	0.096	0.098	0.099

4. Sentiment and liquidity

Liquidity has been found particularly important in EMs. (Chuhan, 1994) shows that poor liquidity is one of the main impediments to foreign institutional investment in EMs. Baker and Wurgler (2006) claim that sentiment-driven mispricing is the result of arbitrage limits. Furthermore, illiquidity imposes impediments on arbitrage opportunities and makes arbitrage positions very costly (Brunnermeier and Pedersen, 2005; Han and Lesmond, 2011). Therefore, this paper uses liquidity of EM sovereign bonds to measure limits of arbitrage. This paper investigates whether the liquidity of EM sovereign bonds impacts sentiment effects. Previous literature studies the relationship between liquidity and assets' returns, and find that expected returns on assets increase with their illiquidity (Amihud and Mendelson, 1986, 1991). To proxy for sovereign bonds' liquidity, I use the bid-ask spreads of EM sovereign bonds (Bekaert, Harvey, and Lundblad, 2007; Favero, Pagano, and Thadden, 2010).

In column (1) of Table 10, I run a regression with sentiment interacted with the bid-ask spread. The estimated coefficient on the interaction term is negative and significant at the 1 percent level, indicating that the impact of sentiment is stronger for EM local sovereign bonds that are less liquid.¹⁹ Economically, a one standard deviation increase in the bid-ask spread is associated with 12.3 basis point increase in EM local sovereign bond returns based on the mean level of sentiment. In column (2), I repeat this analysis for EM external (USD) debt. Similarly, the estimated coefficients on the interaction term remain negative. The slope on the interaction term is significant at the 10 percent level. Economically, a one standard deviation increase in

¹⁹ In column (1) of Table 10, the dependent variables are EM local sovereign bonds in LOC returns since the bid-ask spreads of EM local sovereign bonds are in LOC.

Table 8

Sentiment and future sovereign returns of USD debt, controlling for additional macro variables, The table reports the estimated results in the regressions.

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S^{BW\perp}_{t-1} + \beta_2 Global_t + \beta_3 Country_{i,t} + \varepsilon_{i,t}, \text{ column(1)}$$

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S^{BW\perp}_{t-1} + \beta_2 Global_t + aCP2005_t + \beta_3 Country_{i,t} + \varepsilon_{i,t}, \text{ column(2)}$$

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S^{BW\perp}_{t-1} + \beta_2 Global_t + bLN2009_t + \beta_3 Country_{i,t} + \varepsilon_{i,t}, \text{ column(3)}$$

$$R_{i,t}^{USD} = \alpha_i + \beta_1 S^{BW\perp}_{t-1} + \beta_2 Global_t + cFF6_t + \beta_3 Country_{i,t} + \varepsilon_{i,t}, \text{ column(4)},$$

where $R_{i,t}^{USD}$ is excess return of sovereign bond denominated in USD in month t valued in USD, $S^{BW\perp}_t$ is the level of the orthogonal investor-sentiment index of Baker and Wurgler (2006), $Global_t$, $CP2005_t$, $LN2009_t$, and $FF6_t$ are described in Table 5. $Country_{i,t}$ includes FX (exchange rate return), credit rating, political risk index, the GDP growth rate, the inflation rate, and R_{lag} (one-month lag bond return). t values are calculated using standard errors clustered by country. Country fixed effects and year fixed effects are included. ***, **, and * denote t -statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	(1)	(2)	(3)	(4)
$S_{lag}^{BW\perp}$	-1.174*** (-5.30)	-1.158*** (-5.30)	-1.199*** (-5.40)	-0.794*** (-3.72)
CP2005		-0.201** (-2.25)		
LN2009			0.617*** (8.13)	
Mkt-RF				0.147*** (4.10)
SMB				0.024 (0.74)
HML				0.079*** (3.02)
RMW				0.029 (0.56)
CMA				-0.218*** (-3.36)
UMD				0.043*** (3.82)
FX	-0.370*** (-3.85)	-0.369*** (-3.84)	-0.360*** (-3.77)	-0.337*** (-3.39)
Credit rating	0.006 (0.34)	0.006 (0.36)	0.005 (0.32)	0.006 (0.36)
Political risk index/100	-0.098 (-0.06)	-0.118 (-0.07)	-0.252 (-0.16)	-0.016 (-0.01)
GDP growth	0.083* (1.78)	0.083* (1.78)	0.084* (1.79)	0.084* (1.81)
Inflation	0.037* (2.02)	0.037* (2.02)	0.036* (2.00)	0.036* (1.95)
R_{lag}	-0.030 (-0.98)	-0.031 (-1.01)	-0.027 (-0.88)	-0.019 (-0.64)
VIX	-0.210*** (-6.16)	-0.214*** (-6.14)	-0.178*** (-5.71)	-0.117*** (-4.52)
TED	-0.353 (-1.27)	-0.359 (-1.29)	-0.652** (-2.38)	-0.196 (-0.80)
TERM	-0.830*** (-6.72)	-0.850*** (-6.67)	-0.589*** (-5.21)	-0.835*** (-7.38)
DEF	2.405*** (6.81)	2.491*** (6.71)	1.838*** (5.94)	1.792*** (5.92)
CARRY	0.152*** (4.18)	0.151*** (4.15)	0.154*** (4.22)	0.082*** (3.03)
Constant	0.267 (0.27)	0.619 (0.59)	-0.027 (-0.03)	-1.440 (-1.62)
Observations	4214	4214	4214	4214
R^2	0.244	0.245	0.261	0.262
Adjusted R^2	0.234	0.234	0.251	0.252

the bid-ask spread is associated with 29.5 basis point increase in EM external sovereign bond returns based on the mean level of sentiment. Taken together, the interactions of sentiment and liquidity reported in Table 10 suggest that EM sovereign bonds investors overprice bonds during high sentiment periods and underprice bonds during low sentiment periods, and that the sentiment effects are more pronounced for EM sovereign bonds that are less liquid. Baker and Wurgler (2006) show that sentiment effects are stronger for illiquid stocks because of limits of arbitrage. Similarly, limits of arbitrage are more serious for illiquid EM sovereign bonds, and so sentiment-driven EM sovereign bonds' mispricing is more pronounced.

Table 9

Sentiment and future bond returns of developed markets, This table presents the sentiment effect on monthly sovereign returns (debt issued in local currency) for developed markets. In column (1)–(3), dependent variable is sovereign bond return valued in USD; in column (4)–(6), dependent variable is sovereign bond return valued in LOC. *t* values are calculated using standard errors clustered by country and are reported in parentheses. Country fixed effects and year fixed effects are included. ***, **, and * denote *t*-statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	USD returns			LOC returns		
	(1)	(2)	(3)	(4)	(5)	(6)
$S_{log}^{BW \perp}$	0.422*** (3.58)	0.414*** (3.54)	0.546*** (4.42)	0.099** (2.72)	0.100** (2.75)	0.139*** (3.57)
CP2005		0.187*** (5.09)			−0.0306 (−0.97)	
LN2009			0.553*** (22.26)			0.212*** (13.03)
TED	−1.245*** (−6.14)	−1.230*** (−6.10)	−1.348*** (−6.48)	−0.217*** (−4.06)	−0.219*** (−4.04)	−0.251*** (−4.37)
VIX	0.008 (0.67)	0.009 (0.80)	0.016 (1.31)	0.058*** (14.98)	0.058*** (15.20)	0.0611*** (15.25)
DEF	0.731*** (3.63)	0.599*** (2.90)	0.268 (1.42)	−0.423*** (−5.59)	−0.401*** (−5.10)	−0.596*** (−7.97)
TERM	0.178** (2.73)	0.190*** (2.89)	0.507*** (7.23)	−0.122*** (−4.77)	−0.124*** (−4.70)	0.003 (0.11)
CARRY	10.24** (2.65)	10.42** (2.70)	10.45** (2.69)	−2.378** (−2.16)	−2.409** (−2.20)	−2.300** (−2.09)
Constant	0.594*** (2.81)	0.247 (1.11)	−0.971*** (−4.57)	−0.820*** (−10.22)	−0.763*** (−8.78)	−1.421*** (−14.47)
Observations	6230	6230	6230	6026	6026	6026
<i>R</i> ²	0.066	0.067	0.102	0.093	0.093	0.117
Adjusted <i>R</i> ²	0.058	0.059	0.094	0.085	0.085	0.109

5. Robustness checks

This section provides various robustness checks. First, the possibility that the persistence in the level of sentiment is driving some of the results is discussed. Second, I also repeat previous analyses using alternative sentiment indices.²⁰ Third, I control for the change in the total outstanding amount of EM sovereign bonds. Fourth, I check results regarding to credit ratings. Fifth, I perform weighted regressions. Sixth, I perform subperiod analyses for EMs.

5.1. Changes in sentiment

The results presented so far are based on regressions that have the level of orthogonal BW sentiment as the independent variable. To ensure that the strong statistical significance obtained in the previous sections is not due to the persistence of BW investor sentiment, I estimate regression models using changes in sentiment in Table 11. The estimated models also include pricing factors, local controls, and global controls. I observe from the table that change in sentiment is strongly significant in all the specifications. Other variables related to foreign exchange risk, such as foreign exchange return and CARRY factor, are also highly significant, regardless of the model specification. For pricing factors of CP2005 and LN2009, LN2009 helps to explain EM sovereign bonds denominated in both local currency and USD, while CP2005 only helps to explain EM sovereign returns of external (USD) debt. The *R*²s of the models in Table 11 are also similar to the *R*²s of regressions using sentiment levels.

5.2. Different proxies for investor sentiment

The results presented so far are based on the orthogonal BW sentiment index. This subsection investigates the robustness of the results by different proxies for investor sentiment. Table 12 reports the results of regressing excess returns on the lagged BW composite sentiment index as well as on the pricing factors and macro-related global variables. The estimated results support the hypothesis that EM sovereign bond returns are lower following high sentiment levels.

As an alternative sentiment index I consider the University of Michigan Consumer Sentiment Index (Michigan sentiment index). The Michigan sentiment index is a sentiment measure based on a monthly survey that is mailed to 500 random households and asks their views about both the current and expected business conditions. Lemmon and Portniaguina

²⁰ In unreported regressions, I control for additional macro variables of external debt ratio (a ratio of total external debt to GDP), domestic investment ratio (a ratio of gross domestic investment to GDP), foreign investment ratio (a ratio of foreign direct investment to GDP), and total reserves ratio (a ratio of total reserves (including golds) to GDP). The results of sentiment effects are consistent with Table 7 and Table 8. I also repeat previous analyses controlling for idiosyncratic volatility and economic policy uncertainty (both US EPU and global EPU). The estimated results of sentiment are qualitatively similar and so I will not report them here.

Table 10

Sentiment and future EM sovereign bond returns, controlling for liquidity measure. This table presents the sentiment effect on monthly sovereign returns (debt issued in the local currency) controlling for sovereign bond the bid-ask spreads (Bid_ask_{lag}) and interaction between sentiment and bid-ask spread. In column (1), dependent variable is LOC bond returns of local sovereign debt as bid-ask spreads are in LOC; in column (2), dependent variable is USD bond returns of external (USD) sovereign debt. Country controls in this analysis include FX (the currency devaluation change), credit rating (bond specific rating), GDP growth rate, inflation rate, political risk index, and R_{lag} (one-month lag bond return). Global controls include VIX, TED, TERM, DEF and CARRY. t values are calculated using standard errors clustered by country and are reported in parentheses. Country fixed effects and year fixed effects are included. ***, **, and * denote t -statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	EM sovereign debt issued in LOC (1)	EM sovereign debt issued in USD (2)
$S_{lag}^{BW\perp}$	0.050 (0.18)	−0.997*** (−4.13)
Bid_ask_{lag}	1.709 (1.57)	2.356*** (4.81)
$S_{lag}^{BW\perp} * Bid_ask_{lag}$	−8.116*** (−3.01)	−1.537* (−1.86)
FX	−0.196*** (−5.24)	−0.371*** (−3.80)
Credit rating	0.192 (0.91)	−0.008 (−0.61)
Political risk index	0.021 (0.98)	0.007 (0.39)
GDP growth	−0.022 (−0.73)	0.059 (1.41)
Inflation	−0.050** (−2.49)	0.031* (1.87)
R_{lag}	0.115*** (8.16)	−0.023 (−0.73)
VIX	0.014 (1.07)	−0.197*** (−5.48)
TED	−0.291 (−1.20)	−0.415 (−1.47)
TERM	−0.267*** (−3.17)	−0.911*** (−7.74)
DEF	0.248 (1.20)	2.268*** (6.22)
CARRY	0.065* (2.02)	0.143*** (3.84)
Constant	−2.802 (−1.18)	−0.392 (−0.36)
Observations	2727	4069
R^2	0.142	0.252
Adjusted R^2	0.124	0.241

(2006) document that this measure proxies for investor sentiment. The estimated coefficients on the Michigan sentiment are negative but not significant in Table 13. The weaker results are consistent with the BW indices being better measures of the mood of institutional investors (DeVault, Sias, and Starks 2019).

5.3. Changes in EM sovereign bonds total outstanding amount

In Table 14, I investigate whether investor sentiment affects EM sovereign bond returns through the changes in the EM sovereign bonds total outstanding amount. The results show that the sentiment-return relationship remains. The sentiment effects in EM sovereign bonds are robust after controlling for the change in total outstanding amount.

5.4. Credit ratings

Since average credit ratings vary a lot in EMs, I investigate whether the sentiment-return relationship in EMs is robust to credit ratings. I split the sample into “investment-grade” and “speculative-grade” countries based on their average credit ratings in Table 1. The negative sentiment effects in EM sovereign bonds are robust for the investment-grade sample but not significant for the smaller sample of speculative-grade countries.

Table 11

Sentiment changes and EM sovereign bond returns, This table tests how sentiment changes affect EM sovereign bond returns. $\Delta S_{lag}^{BW\perp}$ is one-month lag of the change in orthogonal BW sentiment. CP2005, and LN2009, are noted in Table 5. FX (exchange rate return), credit rating, political risk index, the GDP growth rate, the inflation rate, and R_{lag} (one-month lag bond return) are local controls. t values are calculated using standard errors clustered by country and are reported in parentheses. Country fixed effects and year fixed effects are included. ***, **, and * denote t -statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	EM sovereign debt issued in LOC						EM sovereign debt issued in USD		
	USD returns			LOC returns			USD returns		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\Delta S_{lag}^{BW\perp}$	-0.659*** (-3.49)	-0.627*** (-3.04)	-0.643*** (-3.35)	-0.672*** (-3.41)	-0.638*** (-2.97)	-0.656*** (-3.24)	-0.846** (-2.28)	-0.806** (-2.25)	-0.728* (-1.99)
CP2005		-0.089 (-1.02)			-0.0899 (-1.14)			-0.202** (-2.30)	
LN2009			0.138** (2.55)			0.161*** (2.80)			0.607*** (8.12)
FX	-1.187*** (-35.49)	-1.186*** (-35.92)	-1.184*** (-35.15)	-0.198*** (-5.65)	-0.197*** (-5.67)	-0.196*** (-5.55)	-0.371*** (-3.89)	-0.370*** (-3.88)	-0.362*** (-3.81)
Credit rating/100	0.237 (0.06)	0.328 (0.08)	0.292 (0.07)	0.571 (0.13)	0.666 (0.15)	0.663 (0.15)	0.607 (0.35)	0.624 (0.36)	0.564 (0.33)
Political risk index/100	0.393 (0.20)	0.357 (0.18)	0.360 (0.19)	1.230 (0.56)	1.190 (0.54)	1.192 (0.54)	0.005 (0.00)	-0.016 (-0.01)	-0.149 (-0.09)
GDP growth	-0.023 (-0.56)	-0.023 (-0.56)	-0.023 (-0.56)	-0.006 (-0.17)	-0.006 (-0.17)	-0.006 (-0.18)	0.082* (1.71)	0.081* (1.71)	0.082* (1.72)
Inflation	-0.019 (-0.73)	-0.019 (-0.73)	-0.019 (-0.72)	-0.048* (-1.95)	-0.049* (-1.94)	-0.048* (-1.94)	0.038** (2.08)	0.038** (2.07)	0.037** (2.06)
R_{lag}	0.056*** (2.80)	0.055*** (2.87)	0.054** (2.73)	0.133*** (4.13)	0.132*** (4.26)	0.133*** (4.16)	-0.024 (-0.81)	-0.025 (-0.84)	-0.021 (-0.71)
VIX	0.019* (1.84)	0.017 (1.40)	0.026** (2.59)	0.006 (0.48)	0.003 (0.24)	0.013 (1.23)	-0.202*** (-6.17)	-0.207*** (-6.15)	-0.171*** (-5.67)
TED	-0.193 (-0.81)	-0.189 (-0.80)	-0.308 (-1.40)	-0.196 (-0.83)	-0.192 (-0.81)	-0.335 (-1.54)	-0.512* (-1.80)	-0.517* (-1.83)	-0.817*** (-2.91)
TERM	-0.167* (-1.87)	-0.171* (-1.97)	-0.108 (-1.11)	-0.203** (-2.42)	-0.208** (-2.54)	-0.132 (-1.42)	-0.647*** (-6.29)	-0.668*** (-6.18)	-0.394*** (-4.34)
DEF	0.392* (2.05)	0.436** (2.13)	0.304 (1.54)	0.307 (1.57)	0.349 (1.69)	0.206 (1.02)	2.501*** (7.17)	2.588*** (7.10)	1.956*** (6.33)
CARRY	0.075** (2.42)	0.075** (2.43)	0.076** (2.43)	0.065* (1.93)	0.065* (1.94)	0.066* (1.97)	0.161*** (4.16)	0.159*** (4.12)	0.162*** (4.18)
Constant	0.918 (0.68)	1.096 (0.76)	0.832 (0.61)	-0.444 (-0.28)	-0.262 (-0.16)	-0.564 (-0.36)	-0.422 (-0.45)	-0.066 (-0.07)	-0.740 (-0.82)
Observations	3167	3167	3167	3027	3027	3027	4200	4200	4200
R^2	0.734	0.734	0.735	0.112	0.113	0.115	0.241	0.242	0.258
Adjusted R^2	0.730	0.730	0.730	0.095	0.096	0.098	0.231	0.232	0.248

Table 12

Robustness checks on alternative sentiment measure, This table tests how alternative sentiment measures affect EM sovereign bond returns. S_{lag}^{BW} is one-month lag of composite investor sentiment used in Baker and Wurgler (2005). CP2005, and LN2009, are noted in Table 5. FX (exchange rate return), credit rating, political risk index, the GDP growth rate, the inflation rate, and R_{lag} (one-month lag bond return) are local controls. t values are calculated using standard errors clustered by country and are reported in parentheses. Country fixed effects and year fixed effects are included. ***, **, and * denote t -statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	EM sovereign debt issued in LOC						EM sovereign debt issued in USD		
	USD returns			LOC returns			USD returns		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
S_{lag}^{BW}	-0.691*** (-2.92)	-0.645** (-2.43)	-0.757*** (-3.38)	-0.772*** (-2.91)	-0.723** (-2.46)	-0.847*** (-3.36)	-1.029*** (-4.66)	-0.967*** (-4.67)	-1.262*** (-5.22)
CP2005		-0.074 (-0.81)			-0.072 (-0.86)			-0.175* (-2.00)	
LN2009			0.153*** (2.97)			0.177*** (3.22)			0.641*** (8.07)
FX	-1.188*** (-35.58)	-1.187*** (-36.01)	-1.185*** (-35.23)	-0.200*** (-5.70)	-0.199*** (-5.73)	-0.197*** (-5.59)	-0.371*** (-3.87)	-0.370*** (-3.87)	-0.360*** (-3.79)
Credit rating/100	-0.008 (-0.00)	0.070 (0.02)	0.050 (0.01)	0.305 (0.07)	0.384 (0.08)	0.402 (0.09)	-2.741 (-0.90)	-2.823 (-0.92)	-2.490 (-0.82)
Political risk index	0.005 (0.24)	0.004 (0.23)	0.004 (0.22)	0.014 (0.61)	0.013 (0.58)	0.013 (0.58)	-0.001 (-0.06)	-0.001 (-0.07)	-0.003 (-0.16)
GDP growth	-0.022 (-0.54)	-0.022 (-0.54)	-0.022 (-0.53)	-0.005 (-0.15)	-0.005 (-0.15)	-0.005 (-0.15)	0.083* (1.78)	0.083* (1.78)	0.083* (1.79)

(continued on next page)

Table 12 (continued)

	EM sovereign debt issued in LOC						EM sovereign debt issued in USD		
	USD returns			LOC returns			USD returns		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Inflation	−0.018 (−0.71)	−0.018 (−0.71)	−0.018 (−0.69)	−0.048* (−1.98)	−0.048* (−1.97)	−0.047* (−1.97)	0.037* (2.02)	0.037* (2.02)	0.036* (2.00)
R_{lag}	0.055** (2.74)	0.054*** (2.81)	0.053** (2.66)	0.132*** (4.03)	0.131*** (4.14)	0.131*** (4.05)	−0.027 (−0.90)	−0.028 (−0.92)	−0.025 (−0.82)
VIX/10	0.017 (1.62)	0.015 (1.27)	0.024** (2.38)	0.004 (0.32)	0.002 (0.14)	0.012 (1.08)	−0.211*** (−6.15)	−0.215*** (−6.12)	−0.179*** (−5.72)
TED	−0.169 (−0.73)	−0.169 (−0.73)	−0.290 (−1.32)	−0.163 (−0.71)	−0.162 (−0.70)	−0.307 (−1.42)	−0.486* (−1.72)	−0.493* (−1.75)	−0.785*** (−2.79)
TERM	−0.249** (−2.32)	−0.245** (−2.26)	−0.196 (−1.70)	−0.302*** (−2.93)	−0.299*** (−2.86)	−0.238** (−2.10)	−0.760*** (−6.61)	−0.769*** (−6.59)	−0.549*** (−5.26)
DEF	0.390* (2.02)	0.427** (2.07)	0.286 (1.45)	0.297 (1.51)	0.332 (1.59)	0.179 (0.89)	2.538*** (6.91)	2.613*** (6.81)	1.942*** (6.06)
CARRY	0.074** (2.52)	0.074** (2.52)	0.075** (2.54)	0.064* (2.04)	0.05* (2.05)	0.066** (2.09)	0.155*** (4.21)	0.154*** (4.20)	0.157*** (4.25)
Constant	1.175 (0.91)	1.301 (0.96)	1.117 (0.87)	−0.110 (−0.07)	0.045 (0.03)	−0.196 (−0.12)	−0.031 (−0.04)	0.239 (0.26)	−0.140 (−0.17)
Observations	3172	3172	3172	3032	3032	3032	4214	4214	4214
R^2	0.734	0.735	0.735	0.113	0.114	0.117	0.242	0.243	0.260
Adjusted R^2	0.730	0.730	0.730	0.097	0.097	0.100	0.232	0.233	0.251

Table 13

Robustness checks on Michigan Sentiment Index, This table tests how the University of Michigan Consumer Sentiment Index affects EM sovereign bond returns. $S_{lag}^{Michigan}$ is one-month lag of Michigan sentiment index. CP2005, and LN2009, are noted in Table 5. FX (exchange rate return), credit rating, political risk index, the GDP growth rate, the inflation rate, and R_{lag} (one-month lag bond return) are local controls. t values are calculated using standard errors clustered by country and are reported in parentheses. Country fixed effects and year fixed effects are included. ***, **, and * denote t -statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	EM sovereign debt issued in LOC						EM sovereign debt issued in USD		
	USD returns			LOC returns			USD returns		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$S_{lag}^{Michigan}/100$	−0.551 (−0.37)	−0.270 (−0.16)	−0.587 (−0.39)	−0.643 (−0.42)	−0.361 (−0.21)	−0.666 (−0.43)	−3.143 (−1.69)	−2.603 (−1.48)	−2.922 (−1.59)
CP2005		−0.095 (−0.97)			−0.095 (−1.07)			−0.182** (−2.28)	
LN2009			0.140** (2.64)			0.163*** (2.86)			0.611*** (8.15)
FX	−1.189*** (−35.44)	−1.187*** (−35.88)	−1.186*** (−35.08)	−0.201*** (−5.70)	−0.199*** (−5.72)	−0.198*** (−5.59)	−0.371*** (−3.88)	−0.370*** (−3.87)	−0.361*** (−3.80)
Credit rating/100	0.034 (0.01)	0.125 (0.03)	0.091 (0.02)	0.361 (0.08)	0.458 (0.10)	0.455 (0.10)	0.623 (0.37)	0.635 (0.38)	0.579 (0.34)
Political risk index	0.005 (0.26)	0.005 (0.24)	0.005 (0.24)	0.014 (0.62)	0.013 (0.59)	0.013 (0.60)	−0.001 (−0.06)	−0.001 (−0.07)	−0.002 (−0.15)
GDP growth	−0.023 (−0.56)	−0.023 (−0.55)	−0.023 (−0.55)	−0.006 (−0.16)	−0.006 (−0.16)	−0.006 (−0.17)	0.083* (1.78)	0.082* (1.78)	0.083* (1.79)
Inflation	−0.018 (−0.70)	−0.018 (−0.70)	−0.018 (−0.69)	−0.047* (−1.96)	−0.048* (−1.95)	−0.047* (−1.94)	0.037* (2.03)	0.037* (2.02)	0.036* (2.01)
R_{lag}	0.056** (2.77)	0.055*** (2.84)	0.054** (2.69)	0.134*** (4.17)	0.132*** (4.32)	0.133*** (4.20)	−0.028 (−0.89)	−0.028 (−0.90)	−0.025 (−0.79)
VIX	0.020* (1.82)	0.017 (1.38)	0.026** (2.57)	0.006 (0.50)	0.003 (0.24)	0.014 (1.24)	−0.207*** (−6.06)	−0.210*** (−6.05)	−0.175*** (−5.59)
TED	−0.244 (−0.95)	−0.235 (−0.90)	−0.361 (−1.50)	−0.251 (−0.98)	−0.240 (−0.92)	−0.391 (−1.65)	−0.581** (−2.06)	−0.580** (−2.06)	−0.880*** (−3.10)
TERM	−0.118 (−1.46)	−0.128 (−1.66)	−0.058 (−0.68)	−0.154* (−2.01)	−0.165** (−2.21)	−0.082 (−1.02)	−0.558*** (−6.72)	−0.581*** (−6.79)	−0.315*** (−3.91)
DEF	0.416** (2.41)	0.471** (2.40)	0.324** (1.89)	0.327* (1.81)	0.381* (1.89)	0.223 (1.24)	2.451*** (6.97)	2.549*** (6.92)	1.902*** (6.08)
CARRY	0.072** (2.35)	0.073** (2.34)	0.073** (2.37)	0.061* (1.90)	0.062* (1.90)	0.063* (1.94)	0.145*** (4.14)	0.146*** (4.15)	0.148*** (4.22)
Constant	1.445 (1.22)	1.315 (1.05)	1.403 (1.16)	0.161 (0.13)	0.038 (0.03)	0.069 (0.06)	3.246 (1.12)	2.944 (1.03)	2.684 (0.94)
Observations	3172	3172	3172	3032	3032	3032	4214	4214	4214
R^2	0.734	0.734	0.734	0.111	0.112	0.114	0.240	0.241	0.257
Adjusted R^2	0.729	0.729	0.730	0.094	0.095	0.097	0.230	0.231	0.247

Table 14

Sentiment and future EM sovereign bond returns, controlling for the change in total outstanding amount. This table presents the sentiment effect on monthly sovereign returns (debt issued in local currency) controlling for the changes in total outstanding amount ($\Delta Totaloutstanding_{lag}$). I divide total market value of bond index by the value price and calculate log change in this total outstanding amount. In column (1)–(3), dependent variable is USD bond returns of local sovereign debt; in column (4)–(6), dependent variable is USD bond returns of external (USD) sovereign debt. Country controls in this analysis include FX (the currency devaluation change), credit rating (bond specific rating), GDP growth rate, inflation rate, political risk index, and R_{lag} (one-month lag bond return). Global controls include VIX, TED, TERM, DEF and CARRY. *t* values are calculated using standard errors clustered by country and are reported in parentheses. Country fixed effects and year fixed effects are included. ***, **, and * denote *t*-statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	<i>EM sovereign debt issued in LOC</i>			<i>EM sovereign debt issued in USD</i>		
	(1)	(2)	(3)	(4)	(5)	(6)
$S_{lag}^{BW \perp}$	−1.139*** (−4.11)	−0.390*** (−3.18)	−0.919** (−2.73)	−1.239*** (−6.07)	−1.173*** (−5.29)	−0.975*** (−3.41)
$\Delta Totaloutstanding_{lag}$	−0.359** (−2.34)	−0.420 (−1.17)	−0.452 (−1.20)	0.050 (1.29)	0.075 (0.84)	0.100 (1.08)
$S_{lag}^{BW \perp} * \Delta Totaloutstanding_{lag}$			0.100 (1.65)			−0.071 (−1.01)
FX		−1.188*** (−35.38)	−1.188*** (−35.38)		−0.370*** (−3.84)	−0.370*** (−3.84)
Credit rating		0.055 (0.55)	0.060 (0.59)		0.006 (0.37)	0.006 (0.36)
Political risk index/100		−0.375 (−0.17)	−0.664 (−0.27)		−0.341 (−0.19)	−0.374 (−0.20)
GDP growth		−0.029 (−0.86)	−0.025 (−0.74)		0.083* (1.77)	0.082* (1.72)
Inflation		−0.026 (−0.95)	−0.019 (−0.79)		0.037** (2.10)	0.038** (2.24)
R_{lag}		0.053*** (2.95)	0.053*** (2.98)		−0.030 (−0.98)	−0.030 (−0.98)
VIX	−0.241*** (−8.21)	0.020* (1.77)	0.020* (1.77)	−0.274*** (−5.65)	−0.210*** (−6.18)	−0.210*** (−6.18)
TED	−0.677 (−1.47)	−0.163 (−0.69)	−0.161 (−0.68)	−0.456 (−1.57)	−0.355 (−1.28)	−0.353 (−1.27)
TERM	−0.343* (−1.98)	−0.210** (−2.42)	−0.209** (−2.41)	−0.748*** (−7.05)	−0.829*** (−6.72)	−0.830*** (−6.72)
DEF	2.058*** (6.25)	0.360* (1.85)	0.362* (1.86)	2.832*** (6.47)	2.409*** (6.84)	2.408*** (6.85)
CARRY	0.516*** (5.92)	0.072** (2.52)	0.072** (2.52)	0.272*** (6.63)	0.152*** (4.18)	0.152*** (4.18)
Constant	5.984*** (3.79)	3.070 (1.32)	3.304 (1.31)	0.738 (1.05)	0.113 (0.12)	0.052 (0.06)
Observations	3172	3172	3172	4232	4214	4214
R^2	0.165	0.735	0.735	0.145	0.244	0.244
Adjusted R^2	0.151	0.730	0.730	0.135	0.234	0.234

5.5. Weighted regressions

The regressions in the models above apply equal weights to all countries. In order to check that the results are not driven by small countries, I adopt three weighting schemes in Table 15. I weight the observations by country population size, total natural resources endowments, and EM sovereign total outstanding amount. The results show that the negative sentiment–return relationship for EM sovereign bonds issued in LOC and USD are robust to all three weighting schemes.

5.6. Subperiod analyses

I also test the role of sentiment on EM sovereign bonds for the post-crisis period (after 2008). The negative sentiment–return relationship is robust for the post-crisis period. The estimated coefficient on sentiment has a larger *t*-statistic for bonds issued in LOC than USD.

6. Conclusion

My research approach focuses on investors' perspective of EM sovereign bond index returns. Changes in EM sovereign bond returns may trigger reweighting of global debt portfolios, which would further affect the cost and the flow of international capital. The analysis of sovereign bond pricing is important for bond portfolio management, risk management and regulation of financial institutions.

Table 15

Sentiment and future EM sovereign bond returns with different weighting schemes. In this table, I repeat the main tests of Table 7 and Table 8 using different weighting schemes. In columns (1)–(3), the dependent variables are the excess USD returns of EM sovereign bonds issued in LOC. I weight the observations by population, total natural resources endowments, and sovereign index total outstanding amount, respectively. In columns (4)–(6), the dependent variables are USD bond returns of external (USD) sovereign debt. *t* values are calculated using standard errors clustered by country and are reported in parentheses. Country fixed effects and year fixed effects are included. ***, **, and * denote *t*-statistics significant at 1 percent, 5 percent, and 10 percent levels, respectively.

	EM sovereign debt issued in LOC			EM sovereign debt issued in USD		
	(1)	(2)	(3)	(4)	(5)	(6)
	Weight: Population	Weight: Total natural resources endowment	Weight: Total outstanding amount	Weight: Population	Weight: Total natural resources endowment	Weight: Total outstanding amount
$S_{lag}^{BW\perp}$	−0.559*** (−3.07)	−0.652*** (−3.02)	−0.737*** (−3.10)	−0.886** (−2.63)	−1.379*** (−3.97)	−1.708*** (−6.10)
FX	−1.232*** (−21.58)	−1.237*** (−26.45)	−1.153*** (−51.56)	−0.503*** (−4.22)	−0.335*** (−4.26)	−0.374*** (−11.63)
Credit rating	−0.093 (−1.50)	−0.128* (−1.72)	0.164*** (2.83)	−0.014 (−1.06)	−0.014* (−1.88)	−0.042 (−1.31)
Political risk index	0.031 (1.36)	0.043 (1.46)	−0.005 (−0.23)	0.011 (0.60)	−0.012 (−0.61)	−0.029 (−1.47)
GDP growth	−0.105*** (−3.49)	−0.108*** (−3.52)	0.006 (0.11)	0.088* (2.01)	0.050** (2.20)	0.016 (0.52)
Inflation	−0.005 (−0.41)	−0.005 (−0.37)	−0.009 (−0.43)	0.049** (2.73)	0.069*** (6.79)	0.048** (2.58)
R_{lag}	0.038* (1.98)	0.044* (1.98)	0.024** (2.66)	−0.022 (−0.51)	−0.013 (−0.37)	−0.057 (−1.07)
VIX	0.035*** (5.79)	0.038** (2.50)	0.024** (2.14)	−0.144** (−2.08)	−0.165*** (−3.82)	−0.146*** (−5.48)
TED	0.098 (0.29)	−0.430 (−0.90)	0.317 (1.17)	−0.259 (−0.87)	−0.382 (−1.34)	−0.067 (−0.18)
TERM	−0.134* (−1.79)	−0.218*** (−2.91)	−0.343*** (−5.90)	−0.780*** (−5.58)	−0.910*** (−6.42)	−1.153*** (−6.78)
DEF	0.230 (0.82)	0.410 (0.95)	0.138 (0.68)	1.881** (2.59)	2.064** (4.53)	1.817*** (7.98)
CARRY	−0.010 (−0.46)	0.022 (0.69)	0.003 (0.13)	0.089 (1.49)	0.152*** (4.20)	0.157*** (3.84)
Constant	−0.255 (−0.13)	−1.722 (−0.80)	−0.695 (−0.44)	−0.545 (−0.67)	−0.407 (−0.34)	1.118 (1.31)
Observations	3172	3172	3172	4214	4214	4214
R^2	0.763	0.815	0.772	0.275	0.225	0.320
Adjusted R^2	0.759	0.812	0.768	0.265	0.215	0.311

This paper studies the links between US equity market sentiment and sovereign bond returns from 30 emerging markets (EM). The empirical analyses of this paper focus on local sovereign debt and external (USD) sovereign debt from 30 emerging countries. This study shows that the investor sentiment index helps to predict EM sovereign bond returns for both local debt and external (USD) debt. Specifically, sentiment is negatively (positively) related to EM (DM) sovereign bond returns, which is consistent with a flight-to-safety phenomenon: when sentiment is low, investors are more pessimistic, prefer safe harbor assets, and require a higher return on risky asset (e.g., EM sovereign bonds). The negative sentiment effects on EM sovereign bonds (both local and external debt) are consistent with EM sovereign bonds being treated as equity-like assets. The negative sentiment-EM sovereign bond returns relationship holds even after I control for a number of other factors including global variables and local variables. Economically, a one standard deviation increase in sentiment is associated with a 43.8 basis point decrease in EM local sovereign bond returns and a 50.8 basis point decrease in external (USD) sovereign bond returns.

Additionally, this paper explores how EM sovereign bond liquidity affects this sentiment-return relationship, and I find that, consistent with larger sentiment effects being driven by limits to arbitrage, investor sentiment effects are stronger for more illiquid bonds. The results suggest that EM sovereign bond illiquidity imposes impediments for arbitrageurs to correct sentiment-driven mispricings.

Appendix A

Table A Credit rating score

This table shows a linear transform of sovereign bond index credit ratings into comprehensive credit ratings from 20 (AAA with positive outlook) to zero (D with default).

Rating Grade	Risk level	Rating	Numerical values
Investment grade	Highest quality	AAA	20
		AA1	19
	High quality	AA2	18
		AA3	17
		A1	16
		A2	15
	Strong payment capacity	A3	14
		BBB1	13
		BBB2	12
		BBB3	11
Speculative grade	Adequate payment capacity	BBB3	11
		BB1	10
		BB2	9
	Likely to fulfill obligations, ongoing uncertainty	BB3	8
		B1	7
		B2	6
		B3	5
	High credit risk	CCC1	4
		CCC2	3
		CCC3	2
		CC	1
	Very high credit risk with possibility of recovery	CC	1
		C	0
	Default	C	0

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